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Agilent Technologies

IC-CAP 2010.08

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IC-CAP Wafer Professional (WaferPro)

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Qt Availability <http://www.qtsoftware.com/downloads>

Patches Applied to Qt can be found in the installation at:

\$HPEESOF_DIR/prod/licenses/thirdparty/qt/patches.

You may also contact Brian Buchanan at Agilent Inc. at brian_buchanan@agilent.com for more information. For details see:

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About IC-CAP Wafer Professional (WaferPro)

Accurate DC/CV (and RF) characterization and statistical modeling of CMOS devices require collecting an extensive amount of measured data from different wafers across several temperatures. DC and RF device modeling teams adopt a modeling flow that includes sophisticated automated measurement solution and efficient data handling capabilities to support advanced statistical analysis and modeling.

Automated measurement software must combine the ability to drive probers, switching matrixes and thermal chucks according to a predefined wafer map, with the ability to run complex DC/CV and RF measurements using a variety of instruments, from Parametric Testers to single box instruments. Furthermore, measured data must be processed upon measurement to calculate Electrical Test (ET) data, such as V_{th} , I_{dmax} , or f_T in the case of RF measurements.

Modern software prober solutions can handle prober positioning, alignment and temperature control very efficiently but lack the flexibility to support a customizable measurement library and complex post process data calculation and analysis. Existing ATE software solutions are typically tailored for the production environment to be efficient for device modeling groups.

IC-CAP's Wafer Professional addresses these short comings and provides a comprehensive and turn-key DC/CV and RF automated characterization solution for the device modeling environment.

What is IC-CAP Wafer Professional?

IC-CAP Wafer Professional (WaferPro) is a measurement application that is fully integrated into the IC-CAP platform. It was designed in partnership with major semiconductor foundries and is a solution to real customer characterization needs. It takes advantage of IC-CAP's powerful measurement and programming environment to enable a library of efficient measurement routines (built-in and user defined), such as adaptive measurement algorithms, that can greatly reduce the overall measurement time. Since the measurement routines are in the IC-CAP environment, either simple or complex post processing, such as calculation of spot measurements or figures of merit, RF de-embedding and direct extraction, can be applied to measured data before data are saved. With Wafer Professional, you will be able to create a simple or as sophisticated a test plan as needed to meet your characterization and modeling measurement requirements. Since it is based on the IC-CAP Modeling Software, it also provides the same efficient control of DC/CV Analyzers, network analyzers, probers, switching matrixes, temperature chucks as well as the powerful 407x and 408x Series of Agilent Parametric Testers.

Major Features

The key features of IC-CAP Wafer Professional include the following:

- Fully Automated measurements across temperature on semi and fully automated probers.
- Real Time Monitoring (user pass/fail condition) Monitor location, status, results and data during measurement execution.
- Support for Agilent Parametric Series 407x/408x ¹ as well as single box instrument benches.
- Device table definition and management.

- Ability to Stop, Pause and Continue test plan.
- Large selection of predefined test DC/CV/RF measurement algorithms, you can also configure new tests and routines in the IC-CAP environment.
- Broadband sweep and Spot Test (ET) measurements and save data to .mdm or .csv files, respectively.
- Data post processing and analysis.
- Ability to run the Test Plan in "Simulation" mode (test plan verification).
- Uses one project to drive different test benches.

¹ Linux only

What's in this Documentation

The goal of this documentation is to help you get started with WaferPro, providing relevant examples that teach you how to use the software, and showing you where you can get more information as you need it. This documentation contains:

- *Before You Begin Using WaferPro* (waferpro) provides the license and system requirements before getting started with WaferPro.
- *WaferPro Project Structure* (waferpro) describes the model files, test plan manager, various project and navigation views, and overview of designing and execution of the test plan.
- *Getting Started with WaferPro* (waferpro) provides steps to guide through the process of starting WaferPro, and the steps to manage the WaferPro projects.
- *WaferPro Project Environment* (waferpro) describes the project work environment, the navigation, test plan, and steps to configure a WaferPro project.
- *About WaferPro Measurement and Routine Files* (waferpro) describes the Measurement Routine and WaferPro routine file, the process to create a routine file in WaferPro. This section also explains the `WPro_lookup_info()` and `WPro_set_info()` PEL functions.
- *About WaferPro Driver File* (waferpro) describes the Driver File which includes the library of PEL drivers for the auxiliary instrumentation.
- *Working with Configuration and Output Files* (waferpro) describes configuration and output files to configure test benches, define Instrument precision options, declare Device Types, probe card dependent pad to pin relation, or spice decks for simulations.

Before You Begin Using WaferPro

Before you begin using WaferPro, you must go through the following initial requirements:

- [License Requirements](#)
- [System Requirements](#)

License Requirements

IC-CAP Wafer Professional (WaferPro) is an add-on product to the IC-CAP Modeling Software platform. Therefore, to use WaferPro, the W8510 WaferPro Module license is required along with the 85199A IC-CAP Software Environment license.

Using WaferPro with a DC/CV Test Bench

Additionally, to use WaferPro with a DC/CV test bench, you will require licenses for the 85199C LCRZ Measurement and the 85199D DC Measurement drivers.

Using WaferPro with an RF Test Bench

To use WaferPro with an RF test bench, you will require the 85199E AC Measurement driver license and the 85199D DC Measurement drivers.

Test Mode Simulation

Optionally, the 85199B IC-CAP Analysis Module license provides you with the capability of utilizing a special *test mode* which will use simulation instead of real time measurements. The simulation mode is useful for debugging newly created test plans.

The following table provides a summary on the part number and their description for the license requirements.

Part Number	Description
85199A	IC-CAP Software Environment
W8510 ¹	IC-CAP Wafer Professional Module
85199C ²	LCRZ Measurement Drivers
85199D ^{2,3}	DC Measurement Drivers
85199E ³	AC Measurement Drivers
85199B ⁴	IC-CAP Analysis Module

¹ Alternatively, the W8511 - IC-CAP Wafer Pro Bundle which includes the 85199A, 85199C, 85199D, and 85199E.

² Required for a DC CV measurement test station.

³ Required for an RF measurement test station.

⁴ Optional, but strongly recommended for running Test Plans in Simulation mode.

System Requirements

The system requirements for IC-CAP Wafer Professional (WaferPro) is described in the following sections.

I/O Interfaces

Instrument I/O is performed via the GPIB interface for the majority of instruments supported by WaferPro. Depending on your PC's operating system, refer to the *Check the Supported Instrument Interfaces* section in *Windows Installation* (winstall) or *Linux Installation* (uinstall) sections for further details on supported interfaces.

A special optical interface is used in conjunction with the Agilent 407x and 408x Series Parametric Testers to control the testhead using a Linux workstation. These interfaces are only available for use with Agilent Parametric testers.

The following table provides the part number and their description for these interfaces.

Part Number	Description
E3160-66504	PCI Optical I/F for 4070 Series
N9170-66502	PCI-Express Optical I/F for 4080 Series

Instrument Library

WaferPro works with all the same test and measurement instrumentation currently supported by IC-CAP. For further details on the instruments supported, refer to *Supported Instruments* (meas).

In addition, WaferPro provides a built-in support for many of the most widely used Probers, Switch matrixes, and Thermal Chucks in today's test environments.

The following table provides a list of supported Wafer Probers, Switching Matrix, and Thermal Controllers.

Wafer Probers	Switching Matrix	Thermal Controllers
Cascade PS21	Agilent HP4070 ¹	Accrettech
Cascade Summit 12K	Keithley 707/708	Cascade Summit
Suss PA300	Agilent HP4080 ¹	Cascade PS21
Accrettech UF3000	Agilent B2200	Temptronic TP032A

¹ Linux System Only.

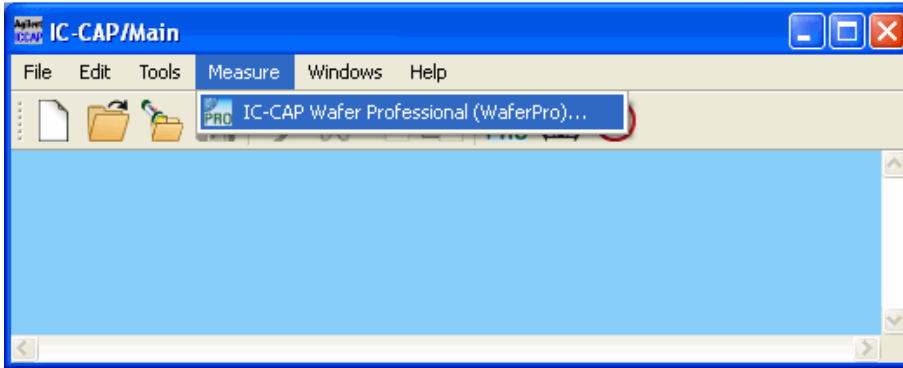
Simulator Requirements

Instead of a direct connection to hardware, a simulator can be used to verify the test plans you create in WaferPro. Current support is provided for spice3 and ADS, however, the test plan simulation mode will work with other spice like simulators (that is, HSPICE, ELDO or Spectre running in spice emulation mode). To configure simulators for use with WaferPro, refer to the *Simulation and Optimization* (sim).

WaferPro Project Structure

The IC-CAP Wafer Professional (WaferPro) is accessible from the IC-CAP Main window **Measure** menu.

IC-CAP Main Window



The WaferPro application's structural design includes two *IC-CAP Model Files* and the *Test Plan Manager*.

IC-CAP Model Files

The IC-CAP model files in WaferPro structural design include the following:

- **Routine model file** - It includes the measurement routine library; with each routine represented by one DUT and its corresponding setups.
- **Driver model file** - It includes the drivers for supported probers, switching matrixes, and thermal chucks.

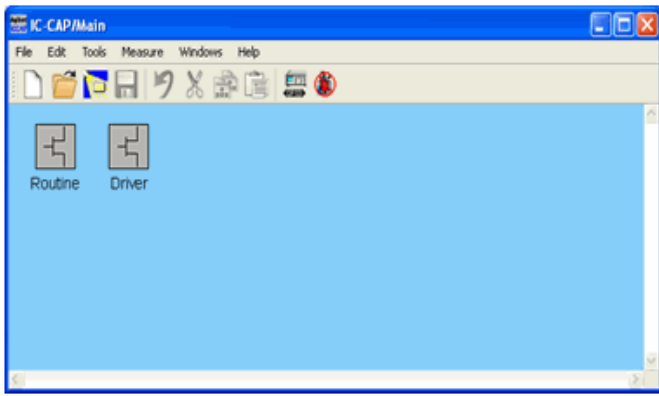
Test Plan Manager

The WaferPro application's structural design includes a Test Plan Manager which runs within the IC-CAP framework.

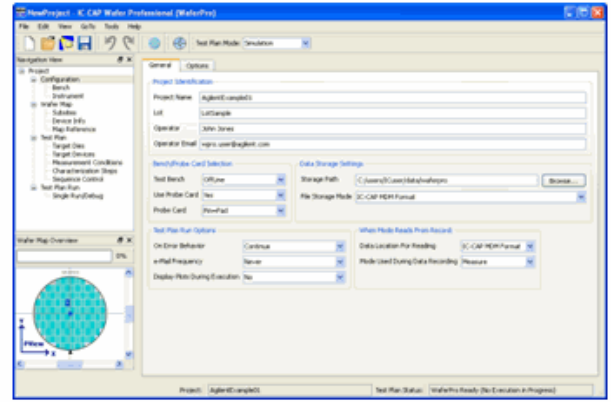
WaferPro communication with IC-CAP

The IC-CAP Wafer Professional (WaferPro) communicates with IC-CAP through an internal link. The following figure shows a simplified flow of communication between WaferPro and IC-CAP.

Communication between IC-CAP and WaferPro



IC-CAP Environment



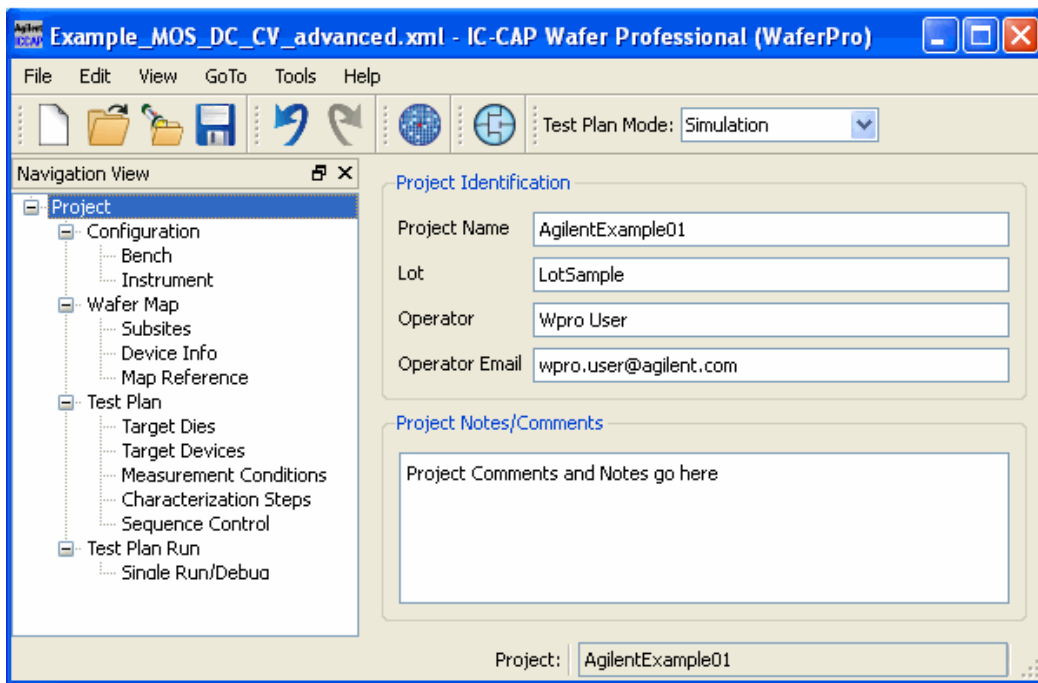
WaferPro Environment

WaferPro Project Window and Navigation Views

Using various Project Window and Navigation Views, you can create a simple or a sophisticated test plan to meet your characterization and modeling measurement requirement.

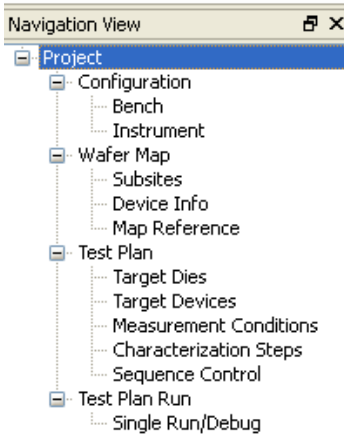
The primary interface to WaferPro is the *Main Project* window as shown in figure below.

Main Project window

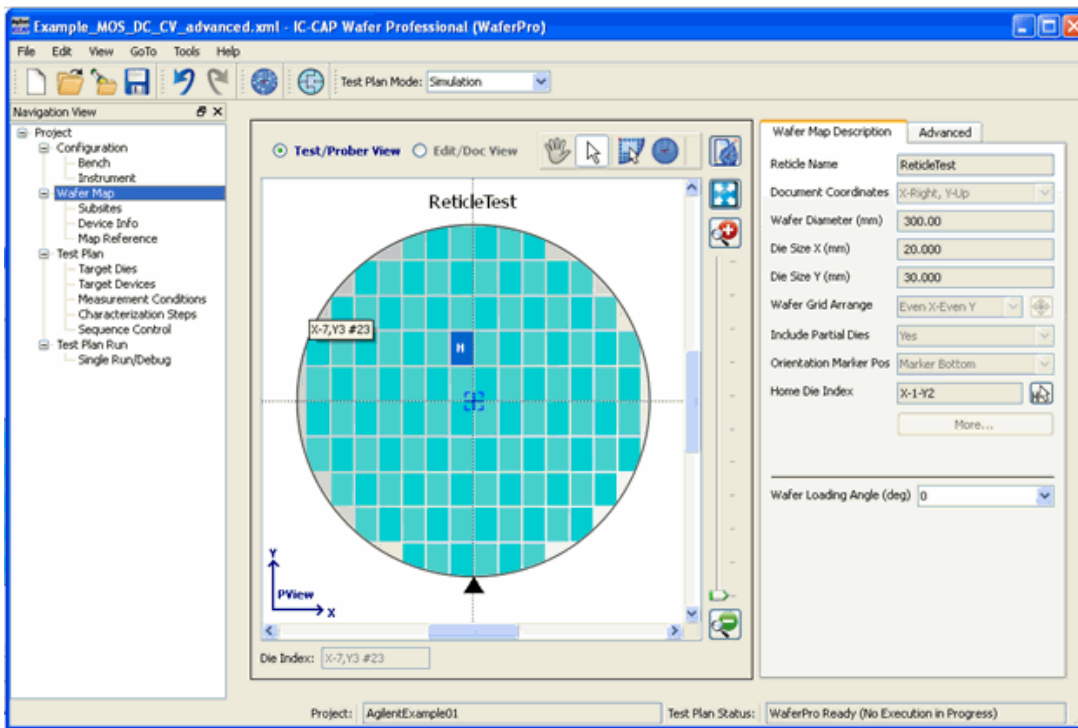


From the *Main Project* window, you can use the *Navigation Tree* (displayed in the below figure) to manage the wafer map information.

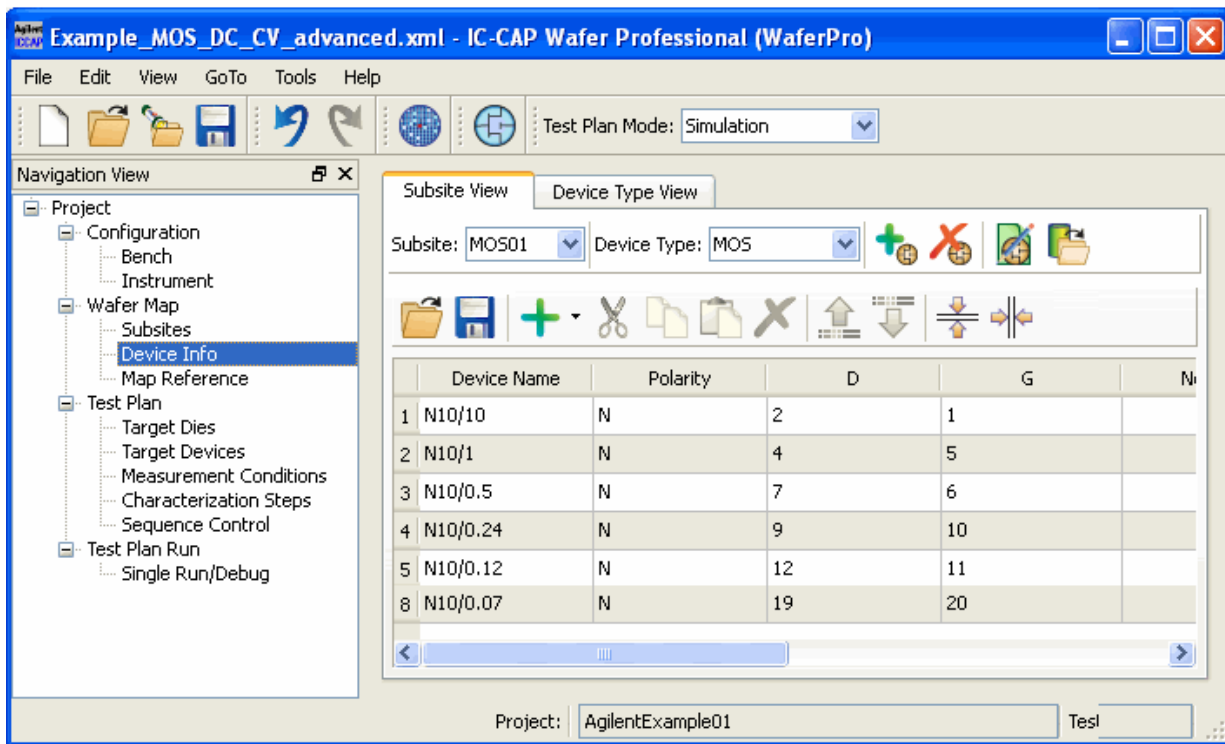
Navigation Tree



Wafer Map Overview

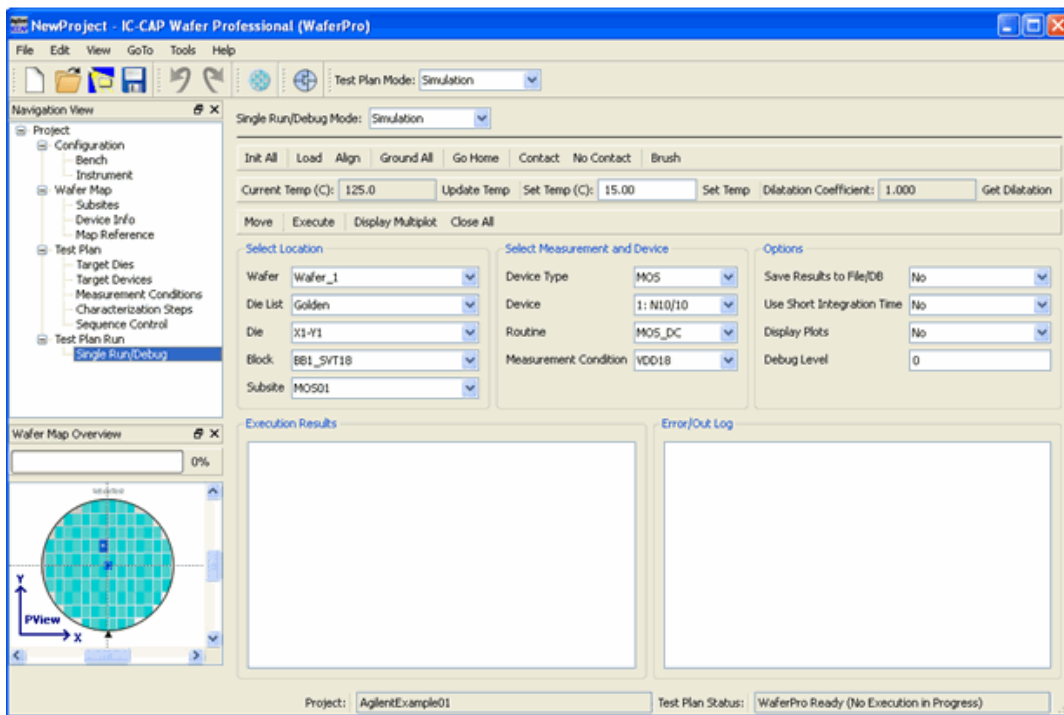


Device Table Information View

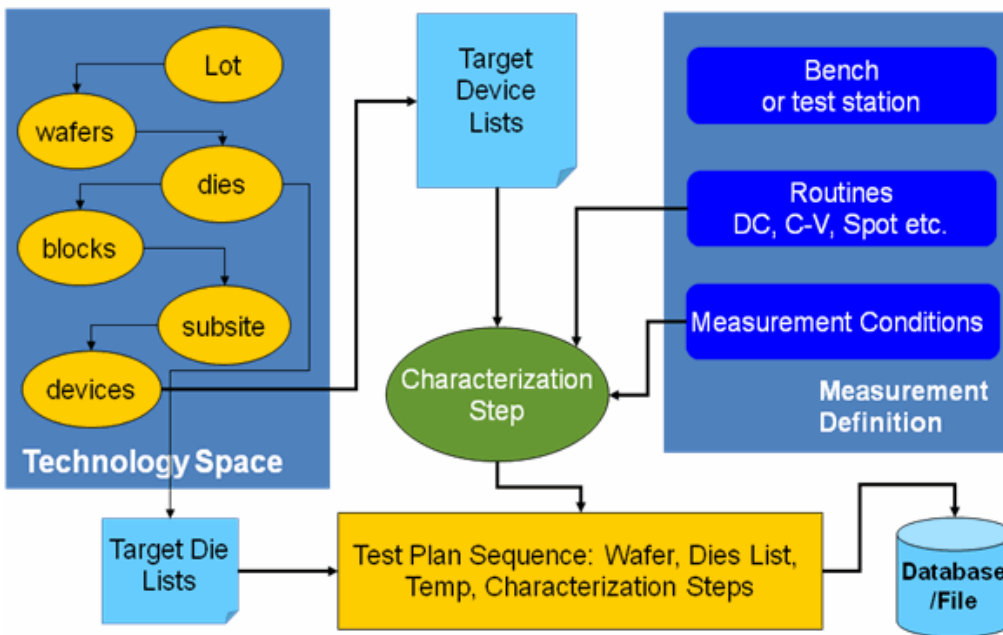


Test Plan Execution View

A Test Plan is defined first by selecting the dies to be measured. The next step is to build lists of *Target Devices* by selecting devices that are suitable to be measured by the same routine algorithm (for example, DC devices, capacitance structures, open devices, etc.). The next step is to build *Characterization Steps* by associating a Routine and its Measurement Conditions (for example, bias Information, integration time, etc.) to each Target Device List. The final Test Plan then lays out the sequence of Characterization Steps applied to specified Wafers, Die List, and Temperature.

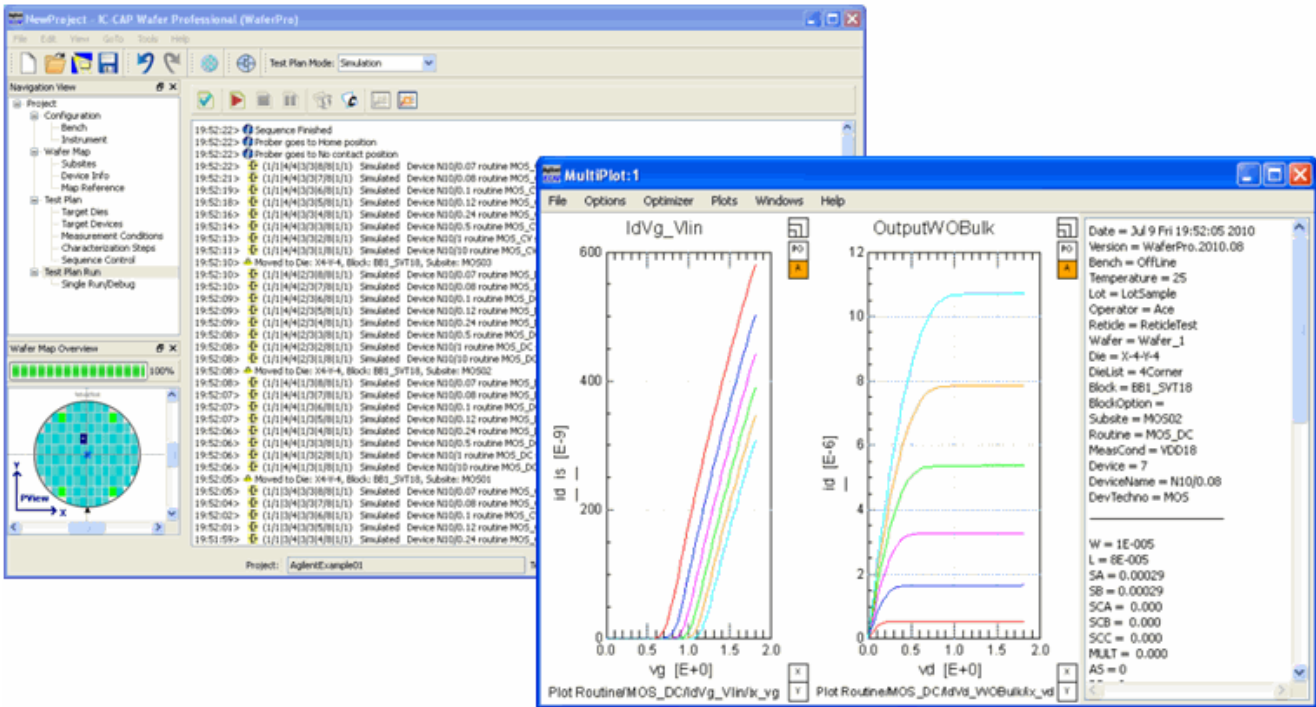


WaferPro Wafer Map and Test Plan Definition Nomenclature



When starting test plan execution, a second IC-CAP batch process running in the background executes the Test Plan. At any time during the execution, you can monitor a comprehensive log which shows the current test plan status and each single test results, including any failed or warning conditions and related comments. It is also possible to view the results of a particular test by simply selecting the specific test and opening the related multi-plot display.

Test Plan Run Page and Log Information with Display Capabilities



Getting Started with WaferPro

Contents

- *Starting WaferPro* (waferpro)
- *Managing WaferPro Projects* (waferpro)
- *An Example WaferPro Session* (waferpro)
- *Preparing For Real Measurements* (waferpro)

An Example WaferPro Session

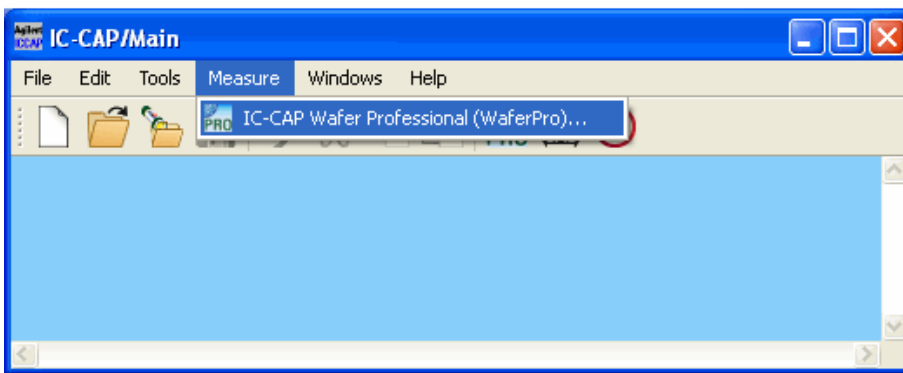
Using WaferPro, you can measure a test project by following this procedure:

- Start the IC-CAP Wafer Professional
- Load an example test project and make a local copy of the example project file.
- Update the Data Storage Path.
- Execute the Test Plan.
- View the Test Plan Log.

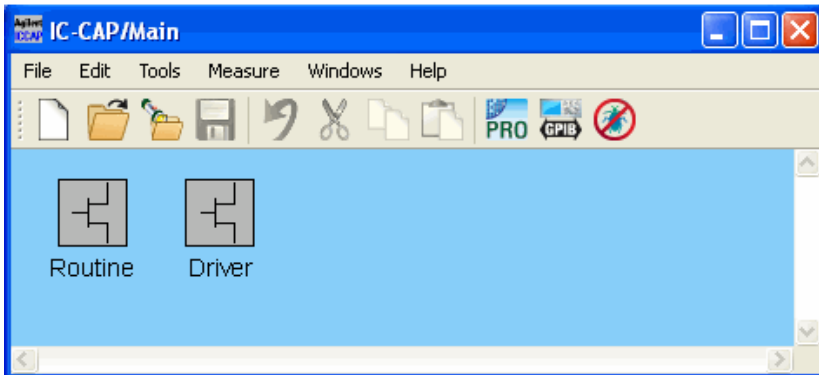
This section provides an example of a WaferPro session, using an example test project.

Starting IC-CAP Wafer Professional

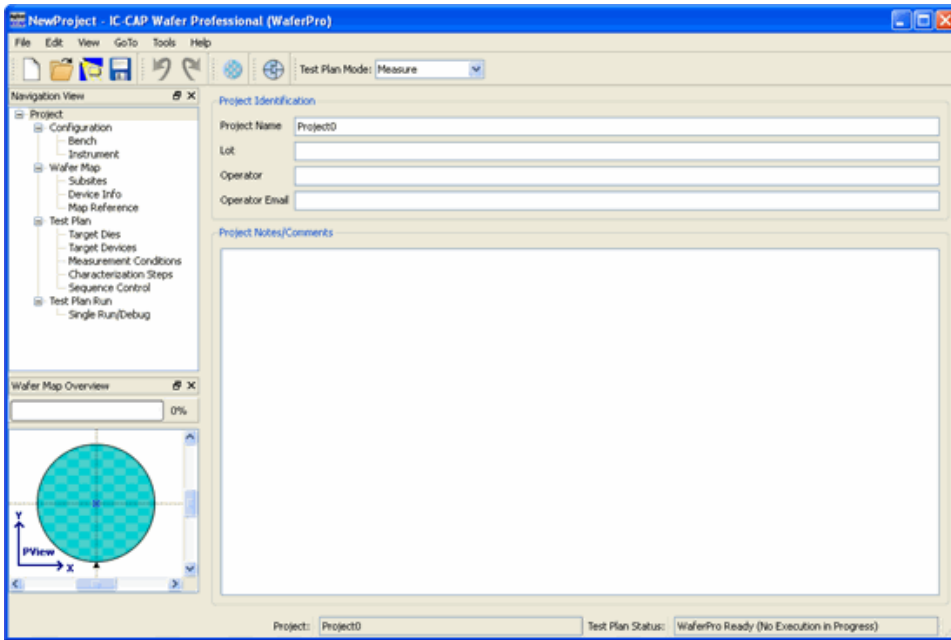
To start IC-CAP Wafer Professional, select **Measure** from the *IC-CAP Main* Window.



The WaferPro Project window appears and the Routine and Driver Model files are loaded.



New Project IC-CAP Wafer Professional (WaferPro) Window



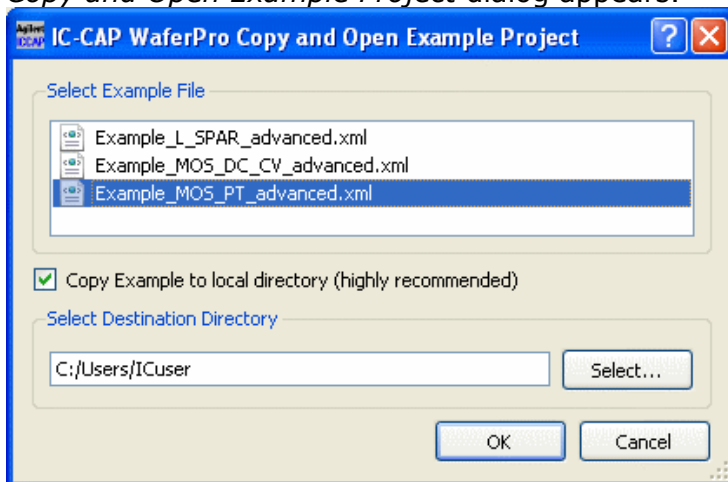
Loading an Example Test Project

An example Test Project, `Example_MOS_PT_advanced`, is provided in the `ICCAP_ROOT/waferpro/examples` directory. Using this example, you can run a *simulated* test plan execution.

By using the simulated test plan mode set in the example, connected the application to the hardware is not required.

To load the example test project:

1. From the WaferPro *New Project* window, select **File > Open Example Project**. The *Copy and Open Example Project* dialog appears.

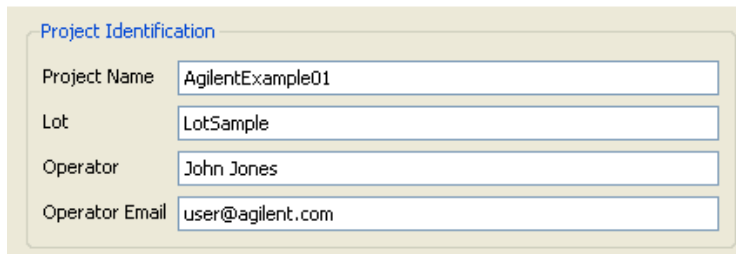


2. In the *Copy and Open Example Project* dialog, select `Example_MOS_PT_advanced` example.
3. Select the **Copy Example to local directory** check box.
4. Set the Destination Directory; click the **Select** button to select a location. By default, the Destination Directory is set to the users \$HOME directory.
5. Click **OK**.

It is recommended that you make a local copy of the example project file by checking the **Copy Example to local directory** check box and setting a destination directory on your local disk. When you click **OK**, a copy of the example is placed in the directory selected

and the example project information is loaded into the WaferPro.

The main WaferPro project window gets updated with the information stored in the example test project for the Project Name, Lot, and Operator.



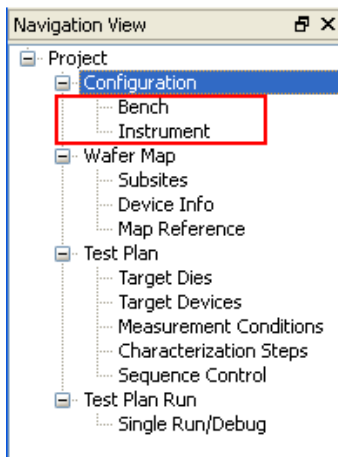
A screenshot of the 'Project Identification' form in the WaferPro application. The form has a light yellow background and a blue title bar. It contains four text input fields with labels to their left: 'Project Name' with the value 'AgilentExample01', 'Lot' with 'LotSample', 'Operator' with 'John Jones', and 'Operator Email' with 'user@agilent.com'.

Updating the Data Storage Path

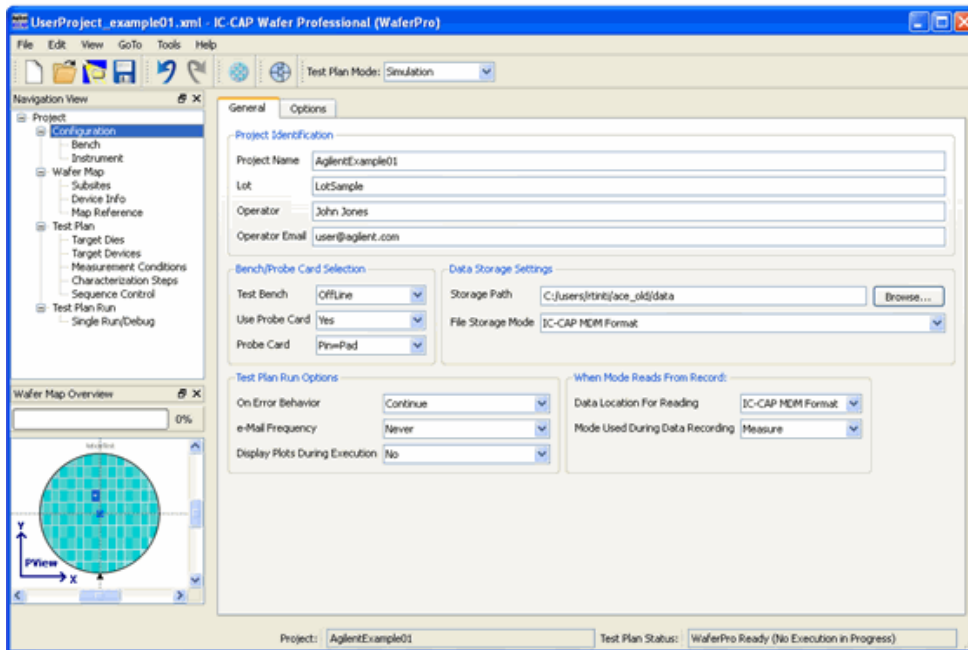
Before moving to the next step, the Data Storage Path needs to be updated to an existing directory on your computer so that WaferPro can properly write the data files during test plan execution. If the Data Storage Path points to an invalid directory, WaferPro will issue error messages in the IC-CAP status window.

To update the Data Storage Path:

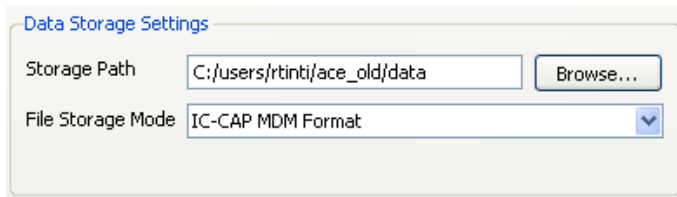
1. In the to the *Navigation View* Tree, select the **Configuration** Node.



The Work area to the right of the Navigation View will display the Configuration View.



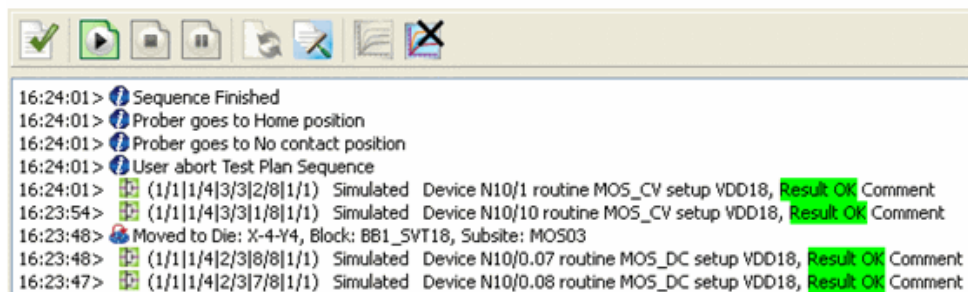
- In the *Data Storage Settings* pane, enter a valid directory path in the **Storage Path** field. You can select **Browse** to select a new data storage directory.



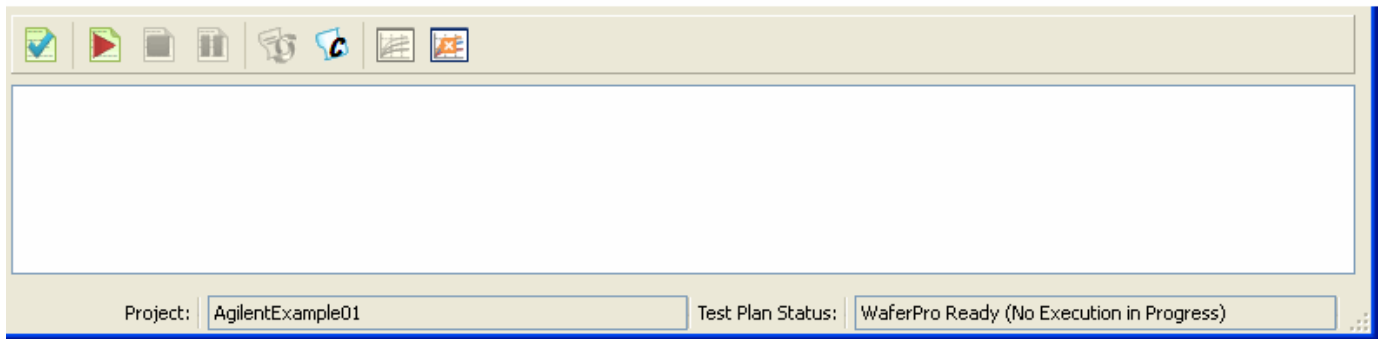
Executing the Test Plan

To execute the test plan:

- In the to the *Navigation View* Tree, select the **Test Plan Run** Node.



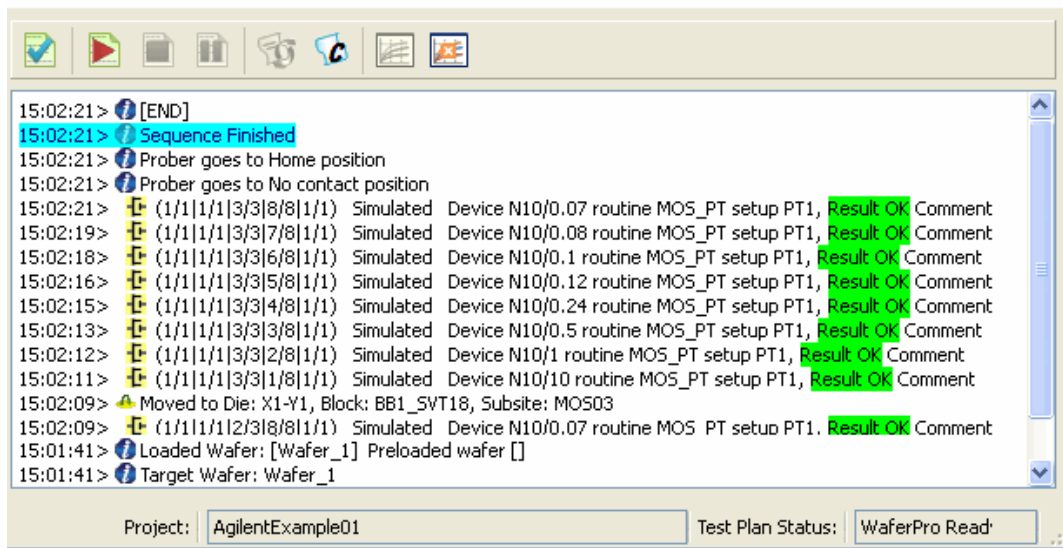
The Work area to the right of the Navigation View will display the Test Plan Run View.



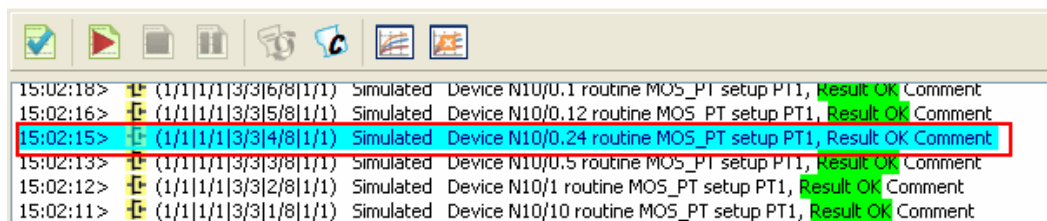
- Click **Start/Resume Test Plan Execution** icon () to start the simulation. You can pause the simulation by clicking the **Stop Test Plan Execution** icon.

Viewing the Test Plan Log

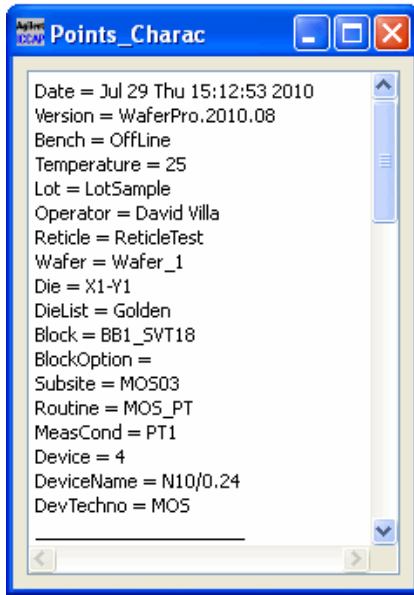
The test plan log is displayed in the log window as shown in the figure below.



Double-click the individual comment to display the details of the selected simulation step.

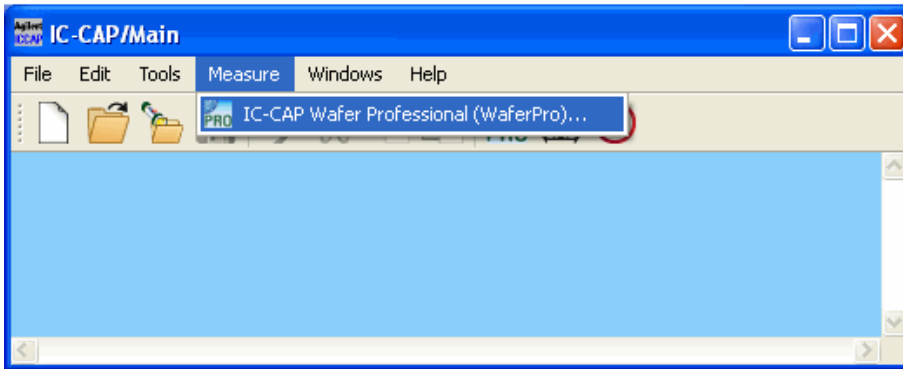


The *Points_Charac* window appears displaying the details of the simulation of the selected device.

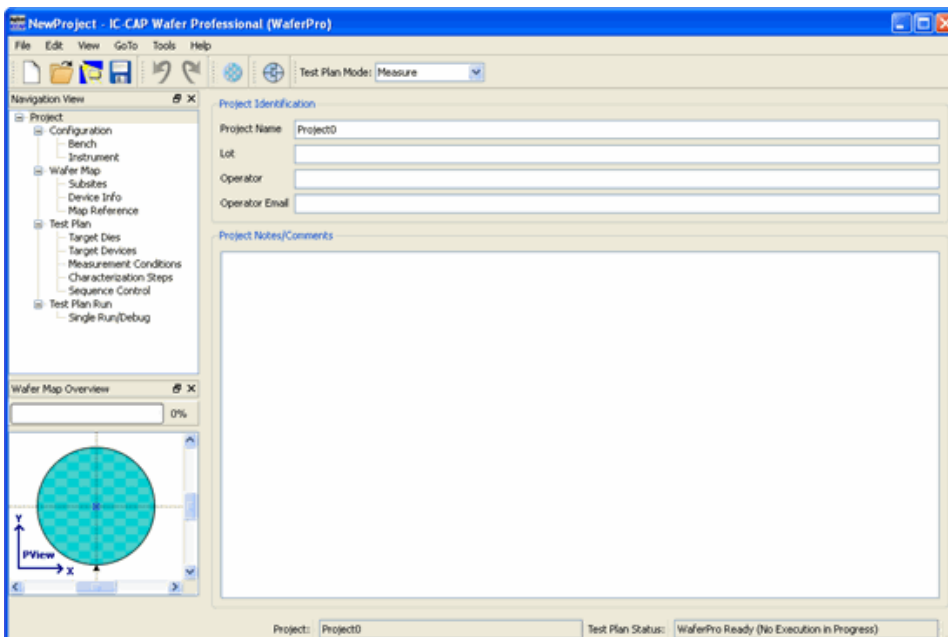
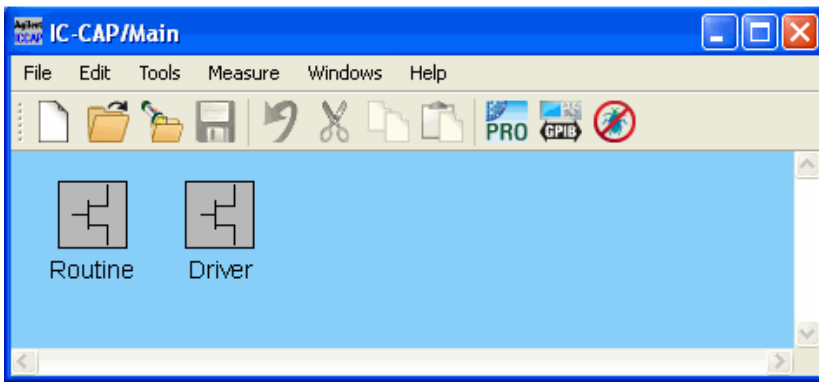


Starting WaferPro

After starting IC-CAP, from the IC-CAP Main Window launch the WaferPro application by selecting **Measure** from the IC-CAP Main window menu.



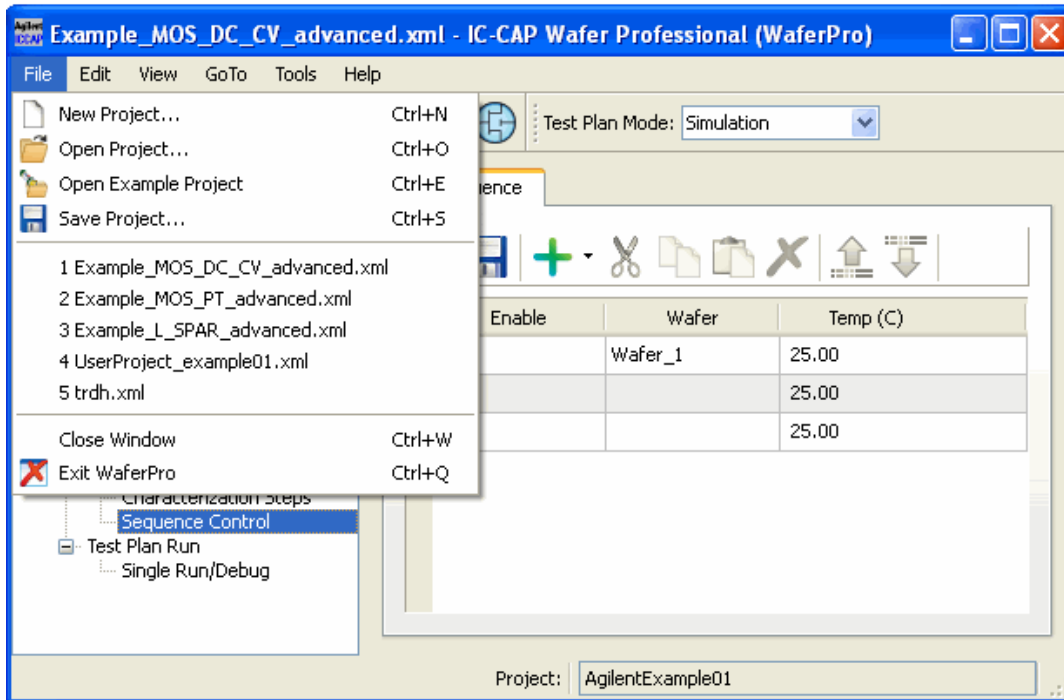
The WaferPro project window appears and the Routine and Driver Model files are loaded.



Managing WaferPro Projects

WaferPro uses projects to meet your characterization and modeling measurement requirements.

Use the **File** menu in the IC-CAP Wafer Professional (WaferPro) Main window to create or open projects.



Creating a New Measurement Project

To create a new project in WaferPro and perform the measurement functions:

1. From the *Wafer Professional (WaferPro)* main window, choose **File > New Project**. A new empty project is initialized.
1. Enter the project details in the *Project Identification* pane.

Project Identification

Project Name	Project0
Lot	
Operator	
Operator Email	

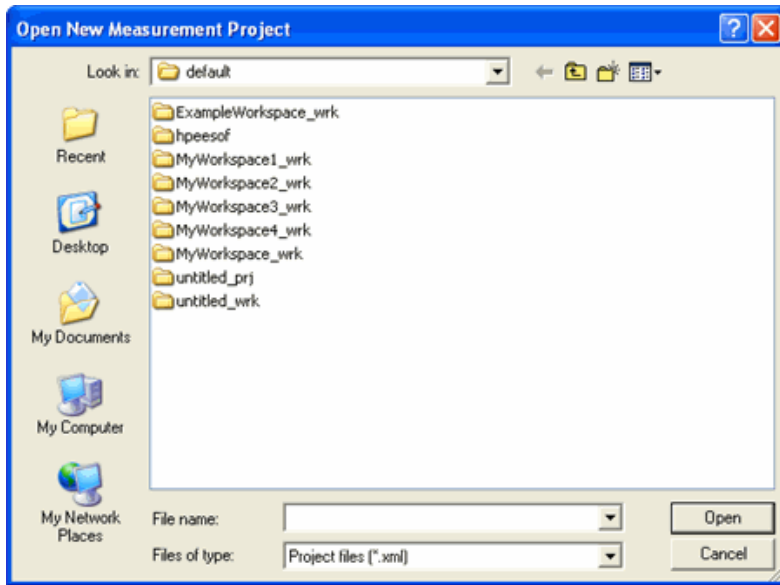
2. Save the project.

Opening an Existing Measurement Project

To reuse and load the existing projects:

1. From the *Wafer Professional (WaferPro)* main window, choose **File > Open Project**.

The *Open New Measurement Project* dialog box appears.



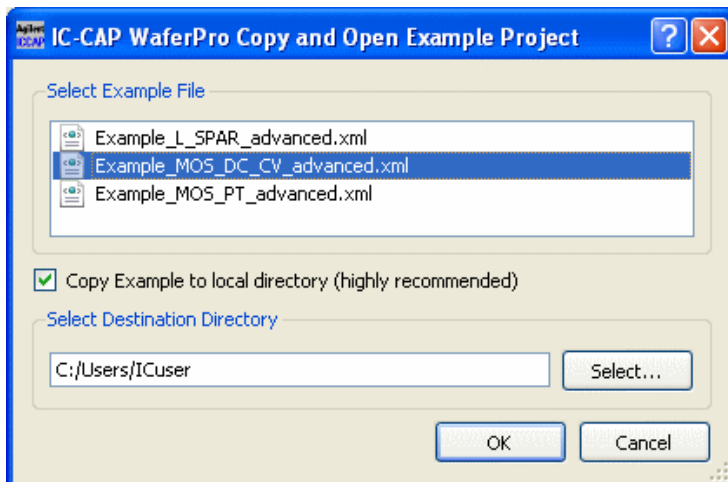
2. Select the project you wish to open and click **Open** to open the project.

Opening an Example Measurement Project

WaferPro provides a set of example projects which demonstrate the various measurement techniques.

To open an example measurement project:

1. From the *Wafer Professional (WaferPro)* main window, choose **File > Open Example Project**. The *IC-CAP Open and Copy Example Project* dialog box appears.

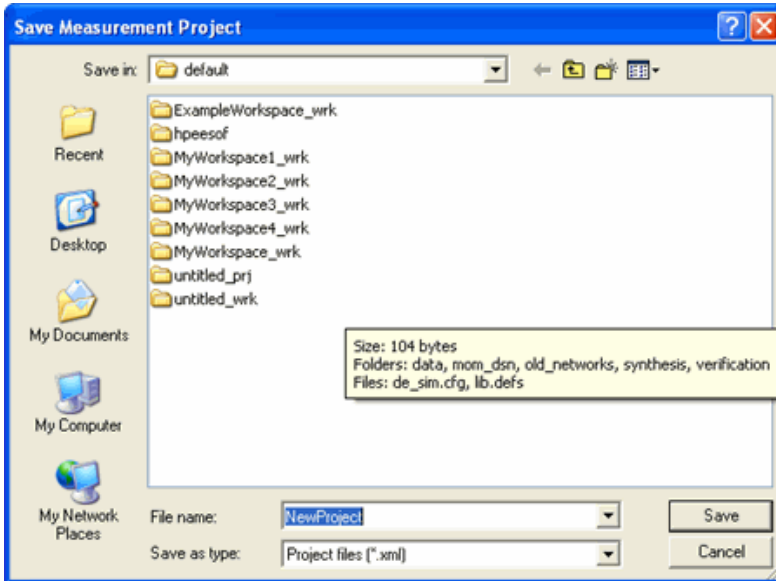


2. Select the example project you wish to open. Check the **Copy Example to Local Directory** option to copy the selected example to a local folder. The **Select Destination Directory** pane displays the default location. Click **Select** to change the default location and select a different location.
3. Click **OK** to open the example project.

Saving a Measurement Project

To save a new project to a specified directory:

1. Choose **File > Save Project** to save the project. The *Save Measurement Project* dialog box is displayed:



2. Type the name of the project and click **Save** to save the project.

Exit from IC-CAP Wafer Professional (WaferPro)

To exit from IC-CAP Wafer Professional (WaferPro):

1. Choose **File > Exit WaferPro**.
2. Click **Yes** to save any unsaved data and exit from WaferPro.

i Exiting from WaferPro also prompts you to save the Routine and Driver files and remove them from the IC-CAP Environment. If you do not wish to exit WaferPro, you can alternatively just close the WaferPro window. In this case, the existing project and the Routine and Driver model files are retained to continue your Wafer Pro Session.

Preparing For Real Measurements

After a walk-through and execution of a simulated test plan, you are ready to make real measurements, but unless you have the same configurations as the default test benches provided with WaferPro, you will need to create your own.

Before adding your own bench to WaferPro, it is advised to make your own working copy of the WaferPro installation to keep the default installation version as a master copy. You also may not have write privileges to the IC-CAP installation directory to be able to modify the files in the installation directory directly as well.

By default, WaferPro finds its configuration and run time files in `$ICCAP_ROOT/waferpro`. Make a copy of this entire directory hierarchy and place the copy in a directory of your choice. After making the copy, you will need to set the environment variable, `ICCAP_WPRO_ROOT`, so that WaferPro can use this new copy instead.

Refer to *Customizing the IC-CAP Environment* (custom) for more details on how to set IC-CAP environment variables.

Adding a New Bench to WaferPro

Step 1: Create a new Benchname.hdw file

1. Launch a new IC-CAP session on the measurement bench you are going to add support for.
2. Open the ICCAP/Hardware Manager window, execute a rebuild and save the IC-CAP hardware configuration file (.hdw) for the measurement instruments in `$ICCAP_WPRO_ROOT/instr`. For additional details on using the IC-CAP Hardware manager and saving the .hdw file, see *Configuring Hardware and Performing Measurements* (meas). It is recommended to use a descriptive name as this filename will be used in the next step for adding your bench to the bench library list as well as what will be displayed in WaferPro.
3. Exit IC-CAP.

Step 2: Update the wproBenchList.cfg file.

The `wproBenchList.cfg` file is located in the `$ICCAP_WPRO_ROOT/config` directory. Open this file in any simple text editor.

 Do not use any word processing software that will embed additional formatting characters. Below is an excerpt from `wproBenchList.cfg`. This excerpt contains the entire definition for the example `SUMMIT_K70x` bench.

```
<SECTION Title="Bench5">
  <VAR Name = "SUMMIT_K70x"></VAR>
  <VAR SlotCount = "1"></VAR>
  <VAR SMUCount = "4"></VAR>
  <VAR CVTisSupport = "FALSE"></VAR>
  <VAR ProberCoordinates = "X-Right, Y-Down"></VAR>
  <VAR GoHomeBeforeDieMove = "FALSE"></VAR>
  <TABLE Title="InstrumentPilot" Format="csv">

    "Type","Name","Controler","Address","CmdSuffix","PinsCount","GROUND","CMHigh","CMLow","SMU1","SMU2",
    "","SMU3","SMU4"
    "Prober","SUMMIT","hplib","27","\r","","","","","","","",""
```

```
"Scanner", "K70xInvert", "hplib", "22", "\r", "6", "9", "7", "8", "1", "2", "3", "4"  
"Chuck", "TP032A", "hplib", "28", "\r", "", "", "", "", "", "", "", ""
```

</SECTION>

The following is a template of the Bench definition. The fields denoted in square brackets [] are the fields that have to be set for the bench being created.

```

1 : <SECTION Title="Bench[#]">
2 : <VAR Name = "[BenchName]"></VAR>
3 : <VAR SlotCount = "[#]"></VAR>
4 : <VAR SMUCount = "[#]"></VAR>
5 : <VAR CVTisSupport = "[Boolean]"></VAR>
6 : <VAR ProberCoordinates = "[ProberCoordinates]"></VAR>
7 : <VAR GoHomeBeforeDieMove = "[Boolean]"></VAR>
8 : <TABLE Title="InstrumentPilot" Format="csv">
9 :
"Type","Name","Controler","Address","CmdSuffix","PinsCount","GROUND","CMHigh","CMLow","SMU1","SMU2
","SMU3","SMU4"
10 :
"Prober","[ProberName]","[InterfaceName]","[GpibAddress]","[TerminationCharacter]","","","","",""
","","",""
11 : "Scanner","[ScannerName]"," [InterfaceName]"," [GpibAddress]","
[TerminationCharacter]","7","9","7","8","1","2","3","4","5"
12 : "Chuck","[TempControllerName]"," [InterfaceName]"," [GpibAddress]","
[TerminationCharacter]","","","","","","",""
13 : </TABLE>
14 : </SECTION>

```

- Line 1: Benches are numbered 1 to N. For a new bench, increment the bench number by 1 from the last bench number used.
- Line 2: Provide a descriptive bench name. The name should be identical to the file name given to the hardware configuration file for the bench.
- Line 3: If the test bench will support cassette loading, set this to the cassette slot count; otherwise set it to 1.
- Line 4: Set # to the number of SMU's your parametric analyzer is configured for.
- Line 5: Set to TRUE if the CV meter can be driven through the TIS interface (407x/8x systems).
- Line 6: Set the X-Y coordinate direction mapping, ie X-Right, Y-Up. This is the prober reference system; most probers will have a X-Right, Y-Up coordinate reference system.
- Line 7: Set to TRUE if prober will return to HOME position before every move; else FALSE. In most cases, leave this setting False. For some probers it may be necessary to reset the reference at every die move in which case this field should be set to True.
- Line 8: No modifications.

- Line 9-12: The instrument pilot table is used to define configuration of context control instruments (prober, chuck, and switching matrix). For each table column, the corresponding instrument value is defined.
 - Type: Describes the instrument family. In WaferPro Driver model file, it corresponds to the instrument driver DUT (Refer to Link to Driver Model file doc).
 - Name: Is the instrument driver name.
 - Controller: If exists, name of the computer GPIB controller.
 - Address: If exists, Instrument GPIB address.
 - CmdSuffix: GPIB command terminator character (that is, \r or \n).

The fields after `CmdSuffix` are variable in number and are meaningful for a switching matrix only. They are used to define the measurement instruments units to switching matrix row connections.

An Example to create a Test Bench to WaferPro

The example bench definition below is created for a test bench having a Summit Prober at address 25, a B2200 Switching matrix at address 20, and a Temperature Controller that will be controlled manually. The parametric tester, in this example case, a B1500 configured with 5 SMUs.

```
<SECTION Title="Bench7">
<VAR Name = "B1500_B2200_HP4284"></VAR>
<VAR SlotCount = "1"></VAR>
<VAR SMUCount = "5"></VAR>
<VAR CVTisSupport = "FALSE"></VAR>
<VAR ProberCoordinates = "X-Right, Y-Up"></VAR>
<VAR GoHomeBeforeDieMove = "FALSE"></VAR>
<TABLE Title="InstrumentPilot" Format="csv">
"Type","Name","Controller","Address","CmdSuffix","PinsCount","GROUND","CMHigh","CMLow","SMU1","SMU2",
", "SMU3", "SMU4", "SMU5"
"Prober","SUMMIT","hpiB","25","\r","","","","","","","","",""
"Scanner","B2200","hpiB","20","\r","7","9","7","8","1","2","3","4","5"
"Chuck","Manual","None","None","","","","","","","","","",""
</TABLE>
</SECTION>
```

Updating the [Precision].iot files.

This step is only required if new measurement instruments are being added to your new bench.

Using the New Bench

After completing the previous steps and putting the new files in place, you should be able to use your new bench.

To use the New Bench:

1. Start IC-CAP and launch WaferPro.
2. Verify that your new bench is available and navigate to the **Bench View** and select the Test Bench from the **Test Bench** list.

Selected Bench Information

Test Bench:	OffLine	
	B1500_B2200_HP4284	
Number of W	OffLine	
Number of A	PA300_K70x	
CV TIS Supp	PS21_4070	
	SUMMIT_K70x	
	SimulationSC01	
	UF3000_4070	
Prober Coordinates	X-Right, Y-Up	

WaferPro Project Environment

Contents

- *Using the WaferPro Project Interface (waferpro)*
- *Understanding the Project View (waferpro)*
- *Configuring a WaferPro Project (waferpro)*
- *Defining the Wafer Layout Specification (Wafer Map) (waferpro)*
- *Block and Subsites Definition (waferpro)*
- *Device Info (waferpro)*
- *Test Plan Definition (waferpro)*
- *Target Dies (waferpro)*
- *Target Devices (waferpro)*
- *Measurement Conditions (waferpro)*
- *Characterization Steps (waferpro)*
- *Sequence Control (waferpro)*
- *Test Plan Run (waferpro)*
- *Single Run and Debug (waferpro)*

Block and Subsites Definition

The Blocks/Subsites definition page allows you to setup and define the wafer blocks and subsites.

The screenshot shows the 'Blocks' and 'Subsites' tabs in the IC-CAP Wafer Professional software. The 'Blocks' tab is active, displaying a table with 5 rows and 2 columns: 'Block Name' and 'Option'. The rows are: 1 BB1_SVT18, 2 BB1_SVT25, 3 NOCONTACT (highlighted), 4 DIO_Block, and 5 Shorted. Below the 'Blocks' tab is the 'Subsites' tab, which is currently inactive. The 'Subsites' tab has a 'Block:' dropdown menu set to 'BB1_SVT18' and a grid icon. Below the dropdown is a toolbar with icons for file operations and editing. The 'Subsites' table has 4 columns: 'Subsite Name', 'X Position', 'Y Position', and 'Notes'. The rows are: 1 MOS01 (X: 2000, Y: 2500), 2 MOS02 (X: 2000, Y: 2700), 3 MOS03 (X: 2000, Y: 2900), and 4 MOS04 (X: 2000, Y: 3100).

Block Name	Option
1 BB1_SVT18	
2 BB1_SVT25	
3 NOCONTACT	
4 DIO_Block	
5 Shorted	

Subsite Name	X Position	Y Position	Notes
1 MOS01	2000	2500	
2 MOS02	2000	2700	
3 MOS03	2000	2900	
4 MOS04	2000	3100	

In the **Blocks** tab, you can create a list of blocks by defining as many block names as needed to describe your wafer floor plan. Each block name must be unique.

i If you do not employ blocks to define your wafer, WaferPro will still need to have at least one block defined to specify the subsites, so you can use the default "whole die" block and then hide the block section of the page using the **Edit block Table** toolbar icon on the *Subsites tab.

From the Subsites tab select a block from the Block drop down list and then define your subsites along with their x-y coordinates on the wafer. Each subsite name defined for a block must be unique. You can continue to create subsite lists for each defined block. The information displayed in the subsite table is block specific.

i All dimensions are in μm .

Subsites Toolbar

The following figure displays the Subsites Toolbar.

The screenshot shows the 'Subsites' toolbar. It includes a 'Block:' dropdown menu set to 'BB1_SVT18' and a grid icon.

The following table describes the various options in Subsites Toolbar:

Button	Name	Description
Block: BB1_SVT18	Select Block	User selectable list of available define blocks.
	Edit Block Table	Toggles between displaying and hiding the block table.

Block and Subsites Toolbar

The following figure displays the Block and Subsites Toolbar.



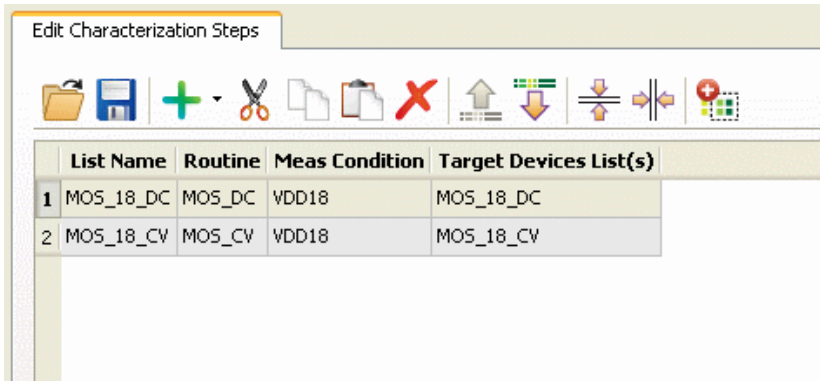
The following table describes the various options in Block and Subsites Toolbar:

Button	Name	Description
	Load from File	The Block or Subsite information table can be manually created or automatically created by loading the information directly from a previously saved file.
	Save to File	Save the Block or Subsites table to a file.
	Add Row	Add a row to the table. Use the pull down menu on this icon to select either to add the row to the end or insert a row above the currently selected row. If no row is selected, an insert will add the row to the end.
	Cut Row	Deletes the entire row and holds the contents in the paste buffer.
	Copy Row	Copies the row contents of the currently selected row to the paste buffer.
	Paste Row	Paste contents of the paste buffer to the currently selected row.
	Delete Row	Deletes the currently selected row.
	Move Row Up	Move currently selected row up in the table.
	Move Row Down	Move currently selected row down in the table.
	Resize All Rows	Resize all rows to fit contents.
	Resize All Columns	Resize all columns to fit contents.

 You can use the right mouse button to bring up a popup menu of the same toolbar icon commands.

Characterization Steps

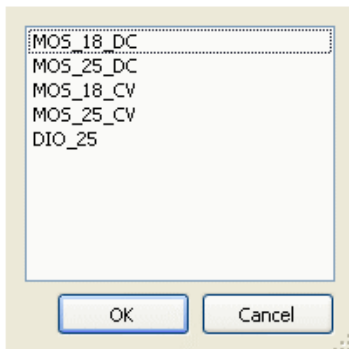
You can specify the Test Plan Characterization Steps you plan to use on this page. For each Characterization Step defined (row) provide a unique name for each in the "List Name" text entry field. Additional for each step specify a Measurement Routine, Measurement Condition for that routine, and the set of devices that will be measured (Target Device List).



The settings for the Routine and Measurement Condition can be manually entered or since they are all user selectable lists, use the mouse pointer to click (right mouse button) in the cell and a drop down list of available values will be presented. Select a value from the list.

These fields will auto complete when manually entering the settings.

The setting for the Target Device Lists field can also be manually entered. Alternatively, you can select the "Select Target Devices from List" toolbar icon to bring up a dialog with a listing of the available Target Device Lists:



You can use the mouse pointer to select one target device from the list. To select multiple values, either hold the **Shift** key to select sequential values or hold down **Ctrl** key to select non-sequential values from the list. When finished, click **OK**.

Characterization Steps Toolbar

The following figure displays the Characterization Steps Toolbar



The following table describes the various options in Characterization Steps Toolbar:

Button	Name	Description
	Load from File	The Characterization Steps information table can be manually created or automatically created by loading the information directly from a previously saved Characterization Steps file.
	Save to File	Save the Characterization Steps table to a file.
	Add Row	Add a row to the table. Use the pull down menu on this icon to select either to add the row to the end or insert a row above the currently selected row. If no row is selected, insert will add the row to the end.
	Cut Row	Deletes the entire row and holds the contents in the paste buffer.
	Copy Row	Copies the row contents of the currently selected row to the paste buffer.
	Paste Row	Paste contents of the paste buffer to the currently selected row.
	Delete Row	Deletes the currently selected row.
	Move Row Up	Move currently selected row up in the table.
	Move Row Down	Move currently selected row down in the table.
	Resize All Rows	Resize all rows to fit contents.
	Resize All Columns	Resize all columns to fit contents.
	Select Target Devices from List.	Opens Select Target Device from List dialog.

Configuring a WaferPro Project

The **Configuration** view provides configuration details of a WaferPro project.

It consists of *General* configuration information tab and an *Options* tab.

General Configuration Tab

The **General** configuration tab contains the following settings:

- Project Identification:**
 - Project Name: AgilentExample01
 - Lot: LotSample
 - Operator: Wpro User
 - Operator Email: wpro.user@agilent.com
- Bench/Probe Card Selection:**
 - Test Bench: OffLine
 - Use Probe Card: No
 - Probe Card: Pin=Pad
- Data Storage Settings:**
 - Storage Path: C:/users/default (with a Browse... button)
 - File Storage Mode: IC-CAP MDM Format
- Test Plan Run Options:**
 - On Error Behavior: Continue
 - e-Mail Frequency: Never
 - Display Plots During Execution: No
- When Mode Reads From Record:**
 - Data Location For Reading: IC-CAP MDM Format
 - Mode Used During Data Recording: Measure

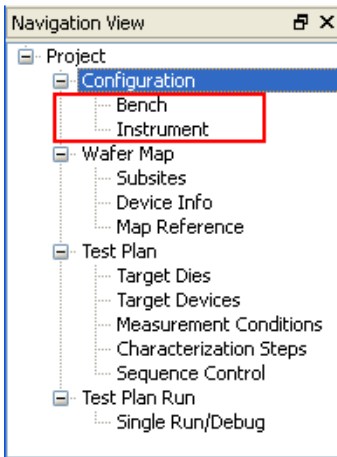
Options Configuration Tab

The **Options** configuration tab contains the following settings:

- Other Test Plan Options:**
 - View Update Interval: 5 sec
 - Test Plan Exe Debug Level: 0
- Simulator Settings:**
 - SIMULATOR: spice3
 - Simulator Type: Spice3 Compatible
- Time Stamp Options:**
 - Add Time Stamp Column in .csv files: No
 - Add Wafers Start/End Times in WaferPro TimeLog.txt file: Yes
 - Add Dies Start/End Times in WaferPro TimeLog.txt file: No

The **Configuration** node in the Navigation View consists of **Bench** and **Instrument** views.

Bench and Instrument Nodes



Configuration View General Tab Options

The *General* tab of Configuration view provides following panes to perform the WaferPro configuration settings:

- [Project Identification](#)
- [Bench/Probe Card Selection](#)
- *Data Storage Settings*
- *Test Plan Run Options*
- *When Mode Reads from Record*

Project Identification

Project Name

Project Name is the text field where you can provide a project name identifier, for example, AgilentExample01.

Lot

Lot is the text field where you can mention the Lot number, for example, LotSample.

 The Lot field value is used as the data storage directory name. This is a required field and should not be left blank or an error will occur during test plan execution.

Operator

Operator is the text field to provide the name of the Operator who will run the test plan, for example, Wpro user.

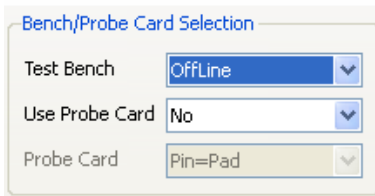
Operator Email

Operator Email is the text field to enter an Operator's email address, for example, wpro.user@agilent.com.

Project Notes/Comments

You can enter any additional information about the current project.

Bench/Probe Card Selection



Bench/Probe Card Selection

Test Bench: OffLine

Use Probe Card: No

Probe Card: Pin=Pad

Test Bench

Test Bench is a user selectable list of test benches. The list of selectable benches is dependent on what is defined in the `WproBenchList.cfg` file.

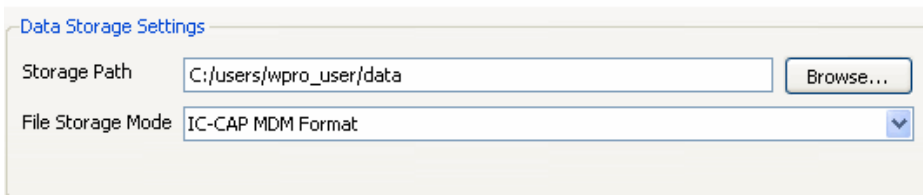
Use Probe Card

If **Yes** is selected (that is, enabled), then the test system is configured with a probe card and you can use it.

Probe Card

Probe Card is a user selectable list to specify the relation of device pad number to the corresponding probe card pin number. A Probe Card file defines one set of the relation (Pad Pin). If no pad-to-pin processing is necessary, select "pad=pin". The selections available in this list are generated from the Probe card Pad to Pin description files in `$ICCAP_WPRO_ROOT/config` directory.

Data Storage Settings



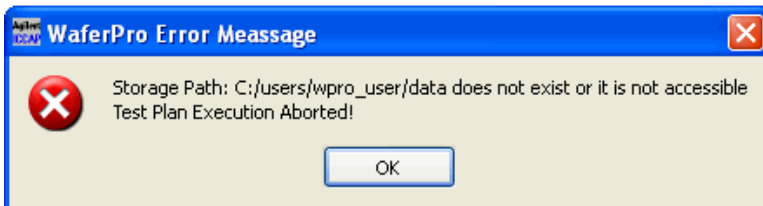
Data Storage Settings

Storage Path: C:/users/wpro_user/data Browse...

File Storage Mode: IC-CAP MDM Format

Storage Path

Set the Storage Path to the directory where WaferPro will store the measured data. By default, the storage path is set to `C:/users/Operator name/data`. Ensure this directory exists before you run the Test Plan, failing which a WaferPro Error Message may appear.

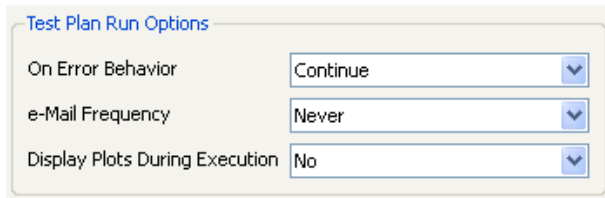


File Storage Mode

File Storage Mode allows you to store the measurement data in **IC-CAP MDM format** or

you may choose not to record the data by selecting **Do not Record data**.

Test Plan Run Options



Test Plan Run Options

On Error Behavior	Continue
e-Mail Frequency	Never
Display Plots During Execution	No

On Error Behavior

On Error Behavior is a user selectable list to specify how WaferPro will behave if a NOK (=Fail) result is produced by a routine applied on a Device. The possible options include Continue, skip device, skip sub-site, skip die, or abort measurement sequence.

eMail Frequency

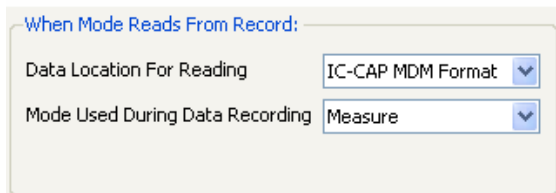
eMail Frequency is a user selectable list to specify the email frequency. The possible options include Never, sequence end, sequence step, or On error and step.

Display Plots During Execution

If **Yes** is selected (that is, enabled), the measurement plot will be displayed during the test plan execution.

 It is recommended to disable this option or the test plan execution may result in a slow performance.

When Mode Reads from Record



When Mode Reads From Record:

Data Location For Reading	IC-CAP MDM Format
Mode Used During Data Recording	Measure

Data Location for Reading

The possible option is **IC-CAP MDM Format** only.

Mode Used During Data Recording

This option is a user selectable list to specify the mode used during data recording. The possible options include Measure, Load from Record, Post Process, or Simulation.

Configuration View Options Tab

The **Options** tab of Configuration view provides following panes to perform the WaferPro options settings:

- [Other Test Plan Options](#)
- [Simulator Settings](#)
- [Time Stamp Options](#)

Other Test Plan Options

Other Test Plan Options

View Update Interval

Test Plan Exe Debug Level

View Update Interval

User Selectable list to specify the time interval Wafer Pro will update the log information during test plan execution

Test Plan Exe Debug Level

Simulator Settings

Simulator Settings

SIMULATOR Simulator Type

Simulator

Specify the simulator to used when Test Plan mode is set to simulation.

 This should be the same key word defined for simulators in the IC-CAP usersimulators file.

Simulator Type

Spice3 compatible simulator or ADS Simulator

Time Stamp Options

Time Stamp Options

Add Time Stamp Column in .csv files

Add Wafers Start/End Times in WaferPro TimeLog.txt file

Add Dies Start/End Times in WaferPro TimeLog.txt file

Add Time Stamp Column in .csv files

If enabled (Yes), will add a time stamp to the measured data .csv files.

Add Wafers Start/End Times in WaferPro TimeLog.txt file

Enable/Disable writing start/end times.

[Add Dies Start/End Times in WaferPro TimeLog.txt file](#)

Enable/Disable writing start/end times.

Bench View

Bench

Selected Bench Information

Test Bench: 

Number of Wafer Slots: Prober Coordinates:

Number of Available SMUs: Go Home Before Die Move:

CV TIS Support:

Cassette/Wafer Table

	Wafer Name	Wafer Option
1	Wafer_1	
5		
6		
12		

Available DC SMU's

	SMU	Driven Nodes
1	SMU1	All
5	SMU5	All
6	SMU6	All

Test Bench

List of test benches to use for measurements.

The specific instrument configuration for the currently selected bench can be viewed by selecting the Show Instrument Configuration icon .

 The Test Bench information is read from the ICCAP_WPRO_ROOT/config/ wproBenchList.cfg file. This configuration file is used to define for Wafer Pro the list of available Test Benches, number of wafer slots, the number of Available SMUs for a specific bench.

Number of Wafer Slots

Non-editable field displaying the number of Wafer slots for the currently selected bench.

Number of Available SMUs

Non-editable field displaying the number of available SMUs for the parametric test instrument used in the currently selected bench.

CV TIS Support

This non-editable field displays whether the currently selected bench supports CV measurements controlled via the TIS.

 CV TIS Support is only valid for test benches configured with an Agilent 407x/408x test system.

Prober Coordinates

These Coordinate Systems are Prober Manufacturer dependent and they are specified in the bench configuration. For example, Cascade Probers are X-Right Y-Up. WaferPro combines Prober Coordinates, Doc Coordinates and the Wafer Loading Angle to determine which the coordinates to be sent to the prober for a move command.

Cassette/Wafer Table

If using an automated prober, the test plan can be applied to several wafers. The cassette/wafer table is used to specify in which slot is located each test wafer. The length of this table is based on the setting for the Number of available Wafer Slots.

Available DC SMU's Table

Lists the available SMUs and and user entry field to specify which SMU can be used to measure a specified node. Possible settings are:

- Left blank to indicate an SMU will not or should not be used for measurements.
- Set to "All" if the SMU is functional and can be used to measure any device node.
- Set to specific nodes if the SMU should be used for. Typically case to use this setting is for SMUs that have weak measurement accuracy or can only force bias.

Instrument View

Instrument

Temperature Control Settings

Long Prober Stabilization (h) 0.000

Short Prober Stabilization (h) 0.000

Wafer Stabilization (min) 0.000

Needle Stabilization (sec) 0.000

Dilatation Method Prober Driven

Dilatation Coefficient %/deg 1.000

Measure during stabilization Yes

Stabilize at First Step No

Instrument Control Settings

Prober

Auto Manual

Chuck Mode

Auto Manual

Switch Matrix Control

Auto Manual

C-V Meter Control

Tis GPIB

Long Prober Stabilization

Specifies the Prober thermal stabilization delay for a large temperature change (more than 60°C) with 4h as typical.

Short Prober Stabilization

Specifies the Prober thermal stabilization delay for a small temperature change (more than 10°C but less than 60°C) with 2h as typical.

Wafer Stabilization

Specifies the Wafer thermal stabilization delay for a wafer change with 20min as typical.

Needle Stabilization

Specifies the needle thermal stabilization delay applied before measure starts after each scribe change.

Dilatation Method

List to choose the method used to take into account wafer thermal dilatation for prober moves. The options are:

- None: No thermal wafer dilatation is taken into account.
- Prober driven: Dilatation factor is computed by Wafer Pro based on prober optical die dimension measurement result.
- User defined: A user-defined dilatation coefficient will be applied using the formula:

$$\text{Move@T} = \text{Move@T=25} * (1 + \text{Coeff} * (T - 25) / 100).$$

Dilatation Coefficient

If Dilatation Method is selected, use this field to specify the dilatation coefficient in %/deg that will be used in the dilatation formula.

Measure During stabilization

If enabled (Yes), during stabilization delay, Wafer Pro will perform measurements and run automated alignment every 5 minutes. If disabled (No), Wafer Pro will wait until stabilization delay time expires before starting measurements.

Stabilize at First Step

If enabled, Yes or No.

 WaferPro prober drivers use index move. As a consequence, wafer dilatation has a weak impact on prober moves.

 "Measure while stabilizing" and "Prober driven" dilatation method must be disabled if the prober using does not support optic index move and automated alignments. (UF300 and UF3000 support "optic based index move" and "automated alignments".)

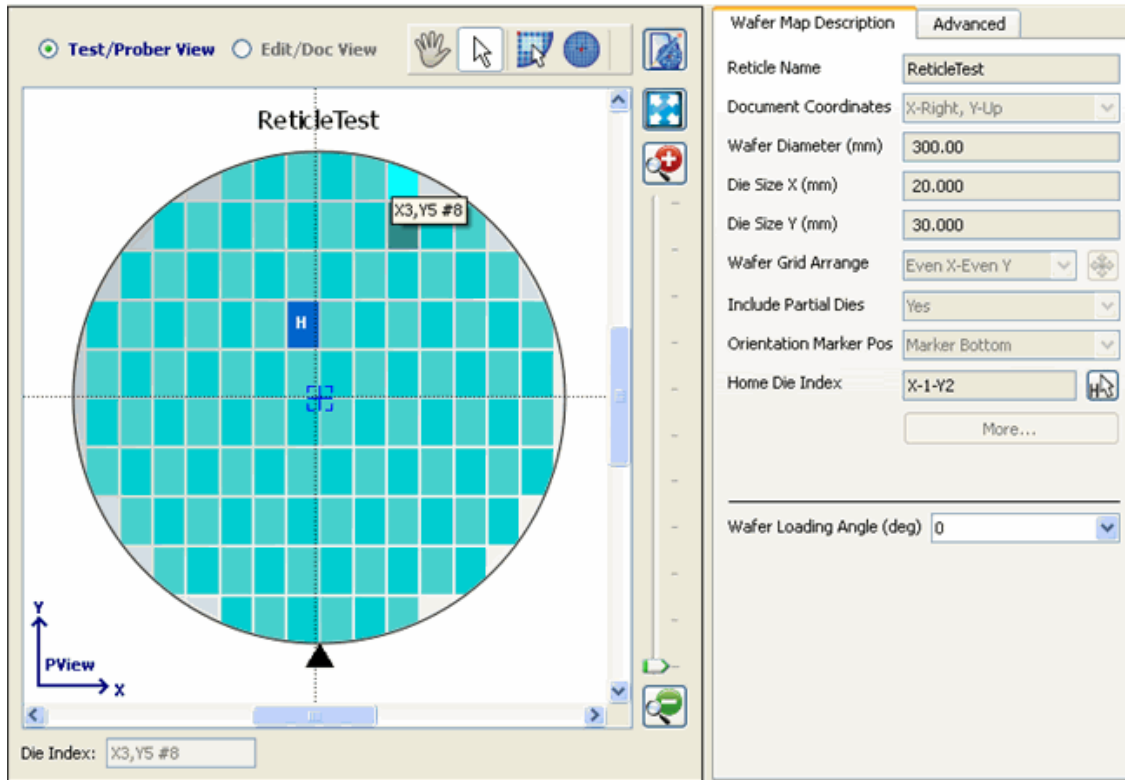
Instrument Control Settings

Instrument Control Settings allows you to set either Auto or Manual control Mode for the Prober, Switching Matrix and Chuck. It also allows the operator to set the CV-Meter control to either the TIS or GPIB. The TIS setting is only for the Agilent 407x/8x family of Parametric testers.

Defining a Wafer Layout Specification

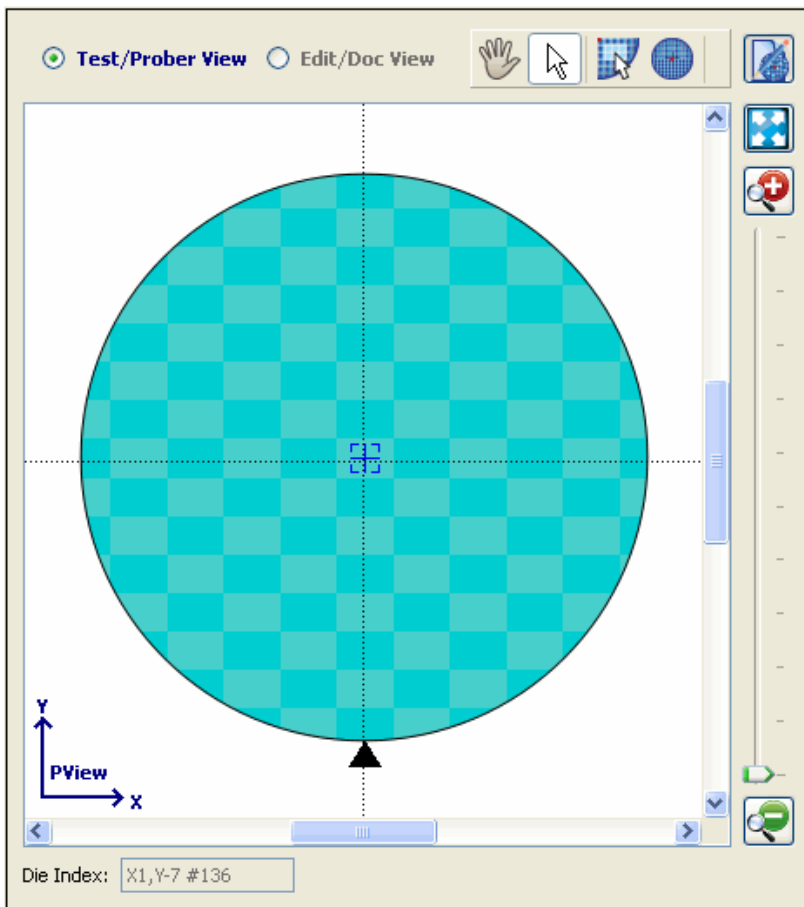
The Wafer Map View is used to describe the wafer layout specification. In this view, you can provide the geographical information that will be used during the characterization sequence.

The data is reticle specific, but can be reused for any characterization run on the same reticle.



Introduction to Wafer Map

The Wafer Map view provides various options to define the Wafer Layout Specification.



Test/Prober View

When selected, the fields in the Wafer Map Description and Advanced Tab are view only. (default)

Edit/Doc View

Select this option to toggle the Wafer Map Description and Advanced Tab field to edit mode.

Wafer Map Toolbar

The following figure displays the Wafer Map Toolbar.



The following table describes the various options Wafer Map Toolbar:

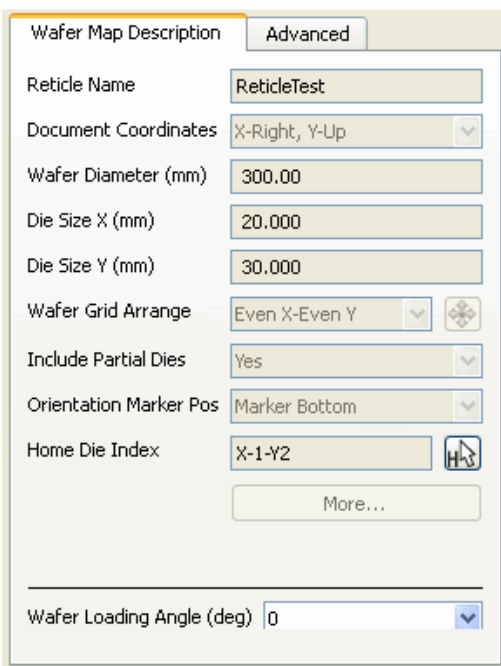
Button	Name	Description
	Hand Tool	Selecting the hand tool, the pointer will turn into a "hand" pointer and allow you to reposition the wafer map inside the viewing area. The hand tool is an alternative to using the scroll bars.
	Selection Tool	Use the selection tool to select die that will you will "deactivate" and remove it from being available when creating the test plan. You can also press and hold the ctrl-key in to select more than one die.
	Deactivate Selected Die	Selecting this icon will deactivate any currently selected die.
	Activate all Die	Selecting this will activate/re-active all die.
	Description/Advanced Tab	Description/Advanced Tab. When hidden, the wafer map view will expand in the tab area.
	Fit Wafer Map to Window	After zooming in or out or repositioning the wafer map in the view area, selecting this tool will re-center and fit the wafer map so it will be displayed in its entirety in the view area.
	Zoom In/Zoom Out	Use this icons and/or the slider bar to Zoom in or Zoom out the wafer map view.

 The Hand Tool is an alternative to using the scroll bars move the wafer map position within the viewing area.

- When using the Selection tool, press and hold the Ctrl-key down to select more than one die.
- Use the Selection Tool to select, then double click to quickly active/deactive die.

Wafer Map Description

The Wafer Map entry fields are by default in Test/Prober View only mode. Select Edit/Doc View mode to edit the fields in this tab.



The image shows the 'Wafer Map Description' dialog box with the 'Advanced' tab selected. The fields are as follows:

- Reticle Name: ReticleTest
- Document Coordinates: X-Right, Y-Up
- Wafer Diameter (mm): 300.00
- Die Size X (mm): 20.000
- Die Size Y (mm): 30.000
- Wafer Grid Arrange: Even X-Even Y
- Include Partial Dies: Yes
- Orientation Marker Pos: Marker Bottom
- Home Die Index: X-1-Y2
- More... button
- Wafer Loading Angle (deg): 0

The following table describes the various fields in Wafer Map Description:

Field	Description
Reticle Name	Text entry field to input a reticle name that will be used in measured data files.
Document Coordinates	List to specify the x/y axis coordinates according to the technology documentation. This may be different than the probe coordinates for a given probe.
Wafer Diameter	Text entry field to specify the Wafer diameter at normal temperatures.
Die Size X	Text entry field to specify the Die Dimension on the X-axis.
Die Size Y	Text entry field to specify the Die Dimension on the Y-axis.
Wafer Grid Arrange	List to describe the die centering on the wafer; ie whether the wafer has odd/even number of die in each row and each column.
Include Partial Dies	If enabled (yes), partial dies will be displayed in the wafer map.
Orientation Marker Pos	List to specify whether the Wafer marker position is Top, Bottom, Left or Right.
Home Die Index	Text entry field to specify the Home Die index. Select the Select Home Die icon next to home index field or the Selection Tool icon on the Wafer Map. Then, use the mouse and click on one of the die in the wafer map to make it the Home die.
Wafer Loading Angle	Specifies the wafer loading angle. Options are 0, 90, 180, and 270.

Wafer Map Advanced View

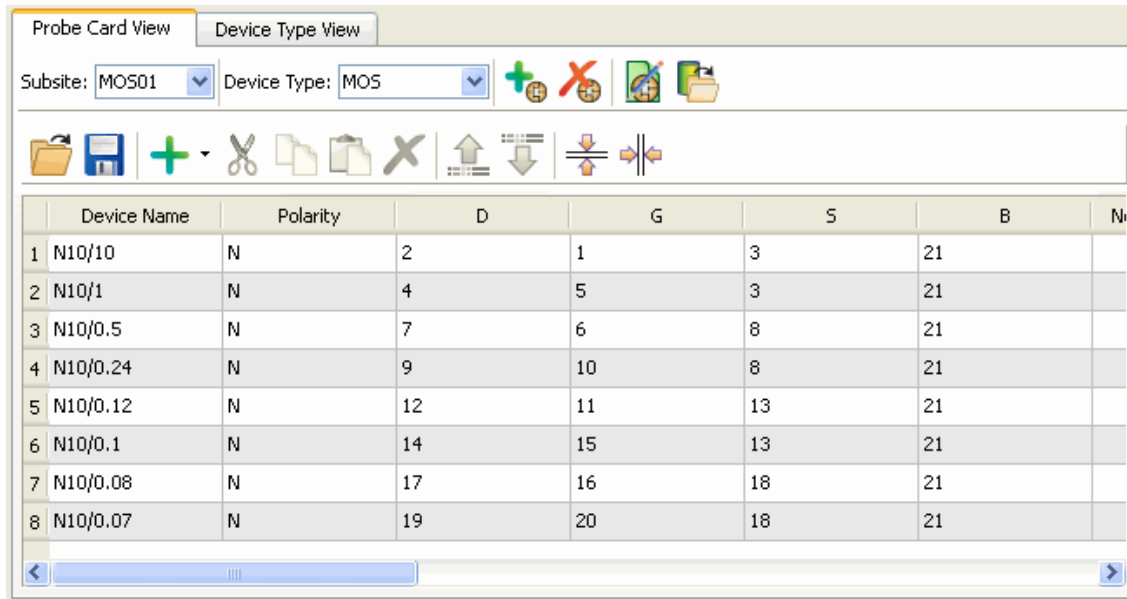
The screenshot shows the 'Wafer Map Description' dialog box with the 'Advanced' tab active. It contains two sections: 'Die ID Numbering' and 'Die Indexing'. In the 'Die ID Numbering' section, 'First Die Location' is set to 'Upper Left', 'Numbering Direction' is 'Horizontal', and 'Alternate Numbering' is 'Yes'. In the 'Die Indexing' section, 'Die Index Modifier dX' is 0 and 'dY' is 0.

The following table describes the various fields in Wafer Map Advanced View:

Field	Description
First Die Location	List to specify the First Die location. Options are Upper Right, Upper Left, Lower Right, and Lower Left.
Numbering Direction	List to specify whether the die numbering sequence is in either the Horizontal or Vertical direction.
Alternate Numbering	If set to Yes, the die numbering, depending on the First Die Location and Numbering direction settings, will alternate the sequence direction with each subsequent row. If set to No, the die numbering sequence will remain in the same direction with each subsequent row. For example if First Die Location = Upper Left and Numbering Direction = Horizontal, then the Die numbering for Alternate Numbering set to Yes will sequence from left to right in the first row, then right to left in the second row, then back to right to left in the next row and so on
Die Index Modifier	Change the x-y coordinates of the X0, Y0 die  Warning Changing the Die Indexing in the middle of a characterization run will break consistency with previously measured data.

Device Info

From the Device Info page you will define and create your device info table. You can manually create the device info table or load it from file.



Subsite View

You can update/edit/view the device info table for the selected Subsite and Device Type. Only the device info for the selected Subsite and device type is displayed in the table.

Subsite View

Device Type View

Subsite: MOS01

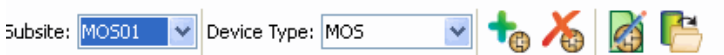
Device Type: MOS



	Device Name	Polarity	D	G	S	B	Node5	Node6	W(um)	L(um)	SA(um)	SB(um)	SCA	SCB	SCC	MULT	AS(um2)	PS(um)	AD(um2)	PD(um)
1	N10/10	N	2	1	3	21			10.00	10.00	290.0m	290.0m	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2	N10/1	N	4	5	3	21			10.00	1.000	290.0m	290.0m	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3	N10/0.5	N	7	6	8	21			10.00	500.0m	290.0m	290.0m	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
4	N10/0.24	N	9	10	8	21			10.00	240.0m	290.0m	290.0m	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
5	N10/0.12	N	12	11	13	21			10.00	120.0m	290.0m	290.0m	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
6	N10/0.1	N	14	15	13	21			10.00	100.0m	290.0m	290.0m	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
7	N10/0.08	N	17	16	18	21			10.00	80.00m	290.0m	290.0m	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
8	N10/0.07	N	19	20	18	21			10.00	70.00m	290.0m	290.0m	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Subsites View First Toolbar

The following figure displays the Subsites View First Toolbar.



The following table describes the various options in Subsites View First Toolbar:

Button	Name	Description
Subsite: MOS01 ▾	Subsite	Select which Subsites device table to display/edit.
Device Type: MOS ▾	Device Type	Select the Device Type table to display for the currently selected subsite.
	Add a Device Type Table to Subsite	Opens the Add Device Type Table dialog from which you can select a Device Type and add a table for it to the currently selected subsite.
	Delete Device Type from Subsite	Opens the Remove Device Type Table dialog from which you can select a Device Type and delete its table from the currently selected subsite.
	Edit Device's # in each Subsite	Displays the Device Table Editor.
	Load Device/Subsites Information from .csv File	Opens the Load Device/Subsites info from .csv file dialog.

Subsites View Second Toolbar

The following figure displays the Subsites View Second Toolbar.

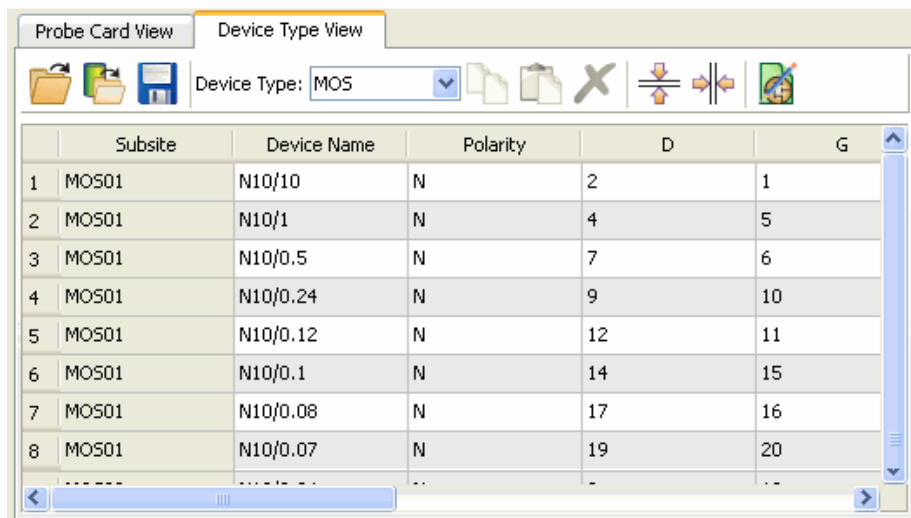


The following table describes the various options in Subsites View Second Toolbar:

Button	Name	Description
	Load from File	The Device Info table can be manually created or automatically created by loading the directly from a previously saved file.
	Save to File	Save the Target Dies table to a file.
	Add Row	Add a row to the table. Use the pull down menu on this icon to select either to add the row to the end or insert a row above the currently selected row. If no row is selected, an insert will add the row to the end.
	Cut Row	Deletes the entire row and holds the contents in the paste buffer.
	Copy Row	Copies the row contents of the currently selected row to the paste buffer.
	Paste Row	Paste contents of the paste buffer to the currently selected row.
	Delete Row	Deletes the currently selected row.
	Move Row Up	Move currently selected row up in the table.
	Move Row Down	Move currently selected row down in the table.
	Resize All Rows	Resize all rows to fit contents.
	Resize All Columns	Resize all columns to fit contents.

Device Type View

You can edit/view the device info table for the selected Device Type from this page. The device info for all defined subsites is displayed in the table.



Device Type View Toolbar

The following figure displays the Device Type View Toolbar.



The following table describes the various options in Device Type View Toolbar:

Button	Name	Description
	Load from File	Load Device info from file to populate the device info table.
	Load Device/Subsites Information from .csv File	Opens the Load Device/Subsites info from .csv file dialog.
	Save to File	Save current the device table info to a file for the selected Device Type.
Device Type: MOS	Device Type	User-selectable drop-down list from which you can select which Device Type's device table to display/edit.
	Copy Selection Contents	Copies the row contents of the currently selected row to the paste buffer when a row is selected. When one or more cells are selected, copies the cell contents to the paste buffer.
	Paste Row/Cell Contents	Paste contents of the paste buffer to the currently selected row or select cells.
	Delete Selected Devices	Deletes the currently selected row.
	Resize All Rows	Resize all rows to fit contents.
	Resize All Columns	Resize all columns to fit contents.

i When using the Load from File toolbar icon from the Device Types View, you will be prompted if you would like to append the loaded device info or to overwrite the current device info table.

Device Table Editor

Device Table Editor is used when manually creating/updated the device info table to specify the number of devices of a certain Device Type for each defined Subsite.

Device Table Editor			
Type to Edit: MOS			
	Subsite	Number of devices	Number of MOS devices
1	MOS01	MOS[8]	8
2	MOS02	MOS[8]	8
3	MOS03	MOS[8]	8
4	MOS04		0
5	JUN01	DIO[0]	0
6	JUN02	DIO[0]	0
7	JUN03	DIO[0]	0
8	W80	DIO[0]	0

Device Table Editor Toolbar

The following figure displays the Device Table Editor Toolbar.



The following table describes the various options in Device Table Editor Toolbar:

Button	Name	Description
Type to Edit: MOS	Type to Edit	Selects the Device Type to display in the device info table.
	Set Devices to Selected Subsites	Sets the number of device for each Device Type for the selected subsites.
	Delete All Devices from Subsite	Deletes all devices of Device Type from the selected subsites.
	Resize All Rows	Resize all rows to fit contents.
	Resize All Columns	Resize all columns to fit contents.

Measurement Conditions

Use the Measurement Conditions page to specify the measurement conditions to apply to the routines you are planning to use in your test plan. Use the Routine drop-down list to select a measurement routine to configure, and then define one or more measurement conditions (column) for that routine. Each Measurement Condition needs a unique name. For each measurement condition define bias conditions and measurement options available for that routine.

Edit Measurement Conditions

Routine: **CONTACT**



	Description
Name	Measurement Condition Name
VRes	(V) Max bias voltage applied for Contact resistance measure
Points	Number of points measured points
Compl	(A) Compliance for all SMUs
AutoCleaningLevel	(ohm) If measured contact resistance exceed the AutoCleaningLevel, the probe brush the needles.
ErrorLevel	(ohm) if measured contact resistance exceed the ErrorLevel, a measurement error is produced.
Resistance_Precision	Precision of the DC instruments for contact resistance measure
AutoRealignment	(Y/N) If resistance exceed Autocleaning level after auto cleaning, then an automated realignment is driven

Measurement conditions also contain special Precision fields which are used to define the instruments precision options for the specified measurement. This fields will have user selectable lists to choose from.

Some routines also contain measurement conditions variables for the Measured Node List (MNL). For each measurement condition defined, you can choose to measure currents on 0 to n terminal nodes where n is defined by the routine. For each MNL, you can enter a comma separated list of nodes. If the MNL field is left empty for that routines setup, then that measurement setup is skipped.

Measurement Conditions Toolbar

The Measurement Conditions Toolbar is shown below:



Button	Name	Description
	Load from File	The Measurement Conditions information table can be manually created or automatically created by loading the information directly from a previously saved Measurement Conditions file.
	Save to File	Save the Measurement Conditions table to a file.
	Add Column	Add a column to the table. Use the pull down menu on this icon to select either to add the column to the end or insert a column to the right of the currently selected column. If no column is selected, insert will add the column to the end.
	Cut Column	Deletes the entire column and holds the contents in the paste buffer.
	Copy Column	Copies the column contents of the currently selected column to the paste buffer
	Paste Column	Paste contents of the paste buffer to the currently selected column.
	Delete Column	Deletes the currently selected column.
	Move Column Left	Move currently selected row up in the table.
	Move Column Right	Move currently selected row down in the table.
	Resize All Rows	Resize all rows to fit contents.
	Resize All Columns	Resize all columns to fit contents.

Sequence Control

The Sequence Control page is the final step in defining the Test Plan. Use this page to define your test Plan sequence. For each sequence step defined (row) specify the Wafer and chuck test temperature setting, the Target Dies (Target Dies List) to test and the Characterization Steps(Characterization Steps List) to run (one or more). You can also enable/disable a defined sequence step for a particular Test Plan execution by setting Yes (enable) or No (disable) in the **Enable** field.

The 'Edit Sequence' dialog box contains a toolbar with icons for file operations and sequence management. Below the toolbar is a table with the following data:

	Enable	Wafer	Temp (C)	Target Dies List	Characterization Steps List(s)
1	Yes	Wafer_1	25.00	4Corner	MOS_18_DC,MOS_18_CV
2	No		25.00		
3	No		25.00		

The settings for Enable, Wafer, Temp, and Target Dies List can be manually entered or since they are all user selectable lists, use the mouse pointer to click (right mouse button) in the cell and a drop=down list of available values will be presented. Select the value for the sequence step.

These above mentioned fields will auto complete when manually entering the settings.

The sequence setting for the Characterization Steps can also be manually entered. Alternatively, you can select the Add Characterization Steps from List toolbar icon to bring up a dialog with a listing of the available Characterization Steps:

The 'Add Characterization Steps' dialog box shows a list of available steps: MOS_18_DC and MOS_18_CV. At the bottom are 'OK' and 'Cancel' buttons.

Use the mouse pointer to select an option from the list. To select multiple values, either hold the Shift key to select sequential values or hold down Ctrl key to select non-sequential values from the list. When finished, click **OK**.

Sequence Control Toolbar

The following figure displays the Sequence Control Toolbar.



The following table describes the various options in Sequence Control Toolbar:

Button	Name	Description
	Load from File	The Sequence Control information table can be manually created or automatically created by loading the information directly from a previously saved Sequence Control file.
	Save to File	Save the Sequence Control table to a file.
	Add Row	Add a row to the table. Use the pull down menu on this icon to select either to add the row to the end or insert a row above the currently selected row. If no row is selected, an insert will add the row to the end.
	Cut Row	Deletes the entire row and holds the contents in the paste buffer.
	Copy Row	Copies the row contents of the currently selected row to the paste buffer.
	Paste Row	Paste contents of the paste buffer to the currently selected row.
	Delete Row	Deletes the currently selected row.
	Move Row Up	Move currently selected row up in the table.
	Move Row Down	Move currently selected row down in the table.
	Resize All Rows	Resize all rows to fit contents.
	Resize All Columns	Resize all columns to fit contents.
	Add Characterization Steps from List	Open the Characterization Steps List dialog with available Characterization Steps.

 You can use the right mouse button to bring up a popup menu of the same toolbar icon commands.

Single Run and Debug

The Single Run/Debug view pane is used to run and test manually all WaferPro Instrument driver capabilities; both measurement instruments and auxiliary instruments. Additionally, this interface can be used to manually run measurements on a single device with a selected Routine and Measurement Condition. Use this page to test newly developed Routines or Measurement Conditions on a test device (versus running a full test plan).

Single Run/Debug Mode

Single Run/Debug Mode is a drop-down list of user-selectable Run and Debug simulation mode.

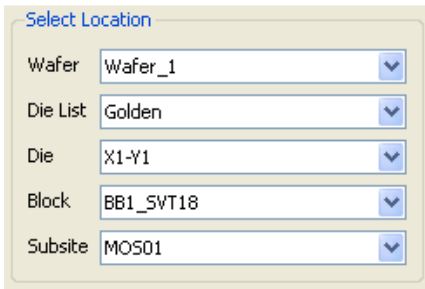
Single Run and Debug Options

The following table describes the various Single Run and Debug Options.

Tab	Description
Init All	Initialize all instruments (run init transform for all drivers).  Execute this command before any other action
Load	Load Wafer (if equipped).
Align	Make the prober run the automated wafer and prober card realignment procedure.
Ground All	Ground all probe card pins.
Go Home	Make the prober move back to its home position.
Contact	Make the prober contact the probe card on the current wafer position.
No Contact	Disconnect the prober card from the wafer.
Brush	Run a prober card brush cleaning.
UpdateTemp	Retrieve the current chuck temperature.
Set Temp	Set the chuck temperature. To do this, set the desired temperature in Celsius in the associated text field then press the "Set Temp" button.
Get Dilatation	Retrieve the wafer dilatation factor at current temperature. To compute the dilatation factor, WaferPro pilots a prober indexed move, then compare the XY distance of the prober move to the user defined die dimension. The ratio gives the current dilatation factor.
Move	Initiate a prober move to the desired location specified in the Select Location fields.
Execute	Execute a measurement routine on a device specified in the Select Measurement and Device fields.
Display Multiplot	Open the data display plot for the select routine.
Close All	Close all open plots.

Select Location Options

Select Location is a group of user-selectable list fields (Wafer, Die List, Die, Block, and Subsite) that specify a particular wafer location to move to when the Move command button is selected.



The screenshot shows a dialog box titled "Select Location" with five dropdown menus:

- Wafer: Wafer_1
- Die List: Golden
- Die: X1-Y1
- Block: BB1_SVT18
- Subsite: MOS01

Select Measurement and Device

Select Measurement and Device is a group of user-selectable lists fields (Device Type, Device, Routine, and Measurement Condition) that specify the device that will be measured when the Execute command button is selected.

Select Measurement and Device

Device Type: MOS

Device: 1: N10/10

Routine: MOS_CV

Measurement Condition: VDD18

Options

Options

Save Results to File/DB: No

Use Short Integration Time: No

Display Plots: No

Debug Level: 0

The following table describes the Options in Single Run and Debug view.

Tab	Description
Save Results to File/DB	If enabled, results are saved to a file.
Use Short Integration Time	If enabled, all Setups will be measured with a precision short, without taking into account the Precision value of the selected setup.
Display Plots	If enabled, this option automatically displays the setups plots of the selected setup during execution.
Debug Level	You can enter the debug level of the test plan.  Note Used during the execution of the single action

Execution Results

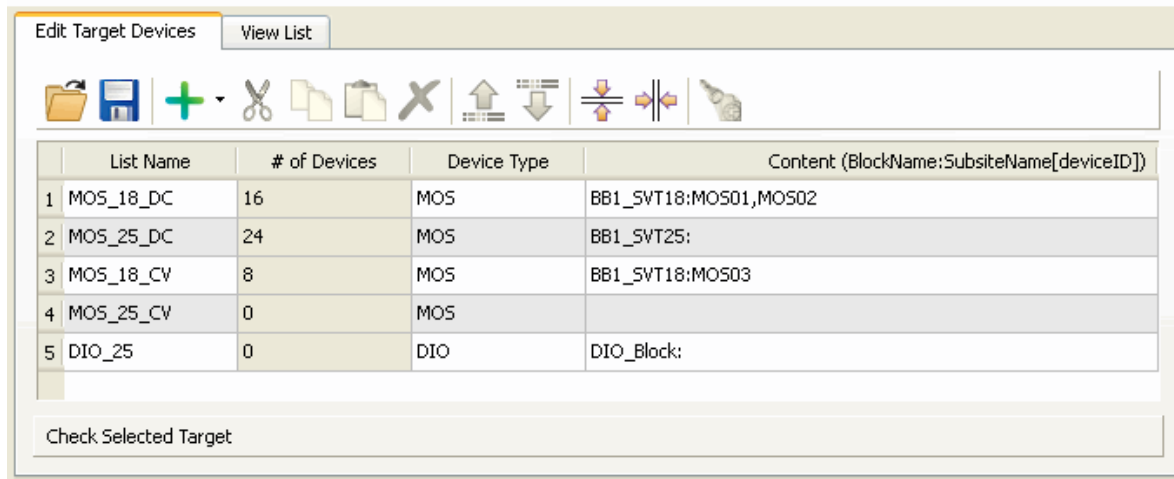
Execution Results window displays the run time status information and results are posted in this text area.

Error/Out Log

Error/Out LogRun window displays the time Errors and Out information are posted in this text area.

Target Devices

The Target Devices page is used to define devices list. Each list represents a set of devices measured with the same measurement routines under the same measurement conditions. The lists should usually group the devices per type and technology options.



Edit Target Devices Toolbar

The following figure displays the Edit Target Devices Toolbar.



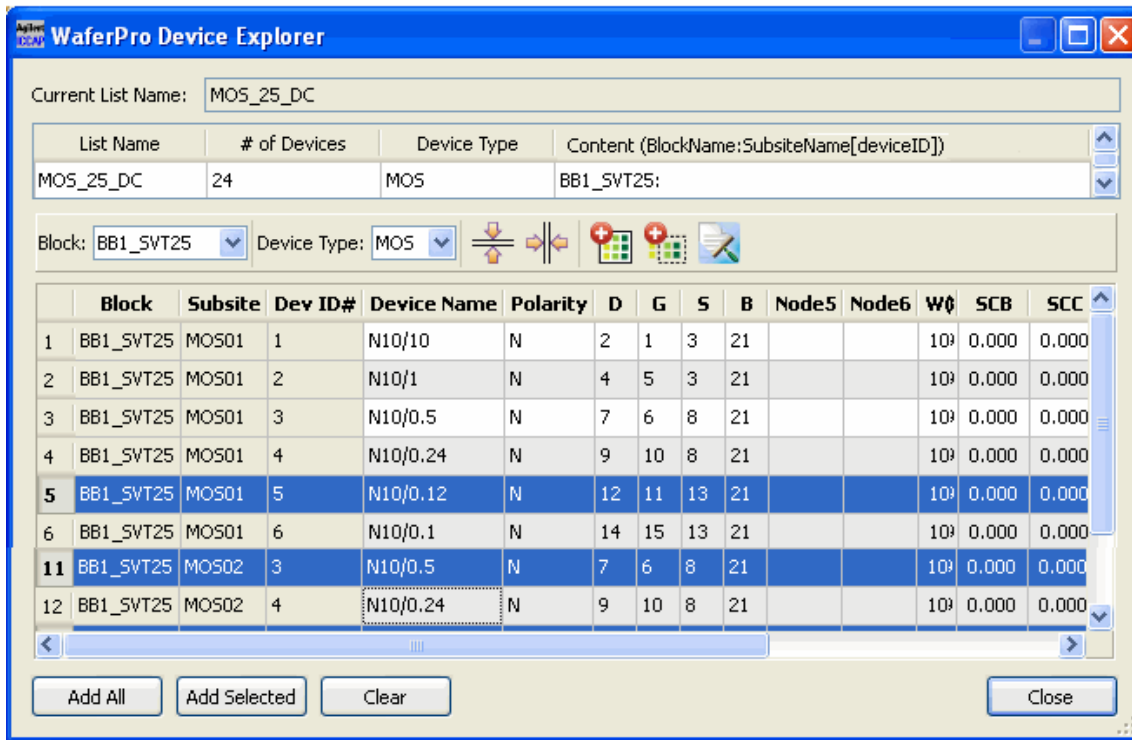
The following table describes the various options in Edit Target Devices Toolbar:

Button	Name	Description
	Load from File	The Target Device table can be manually created or automatically created by loading the information directly from a Device Table file.
	Save to File	Save the Target Devices table to a file.
	Add Row	Add a row to the table. Use the pull down menu on this icon to select either to add the row to the end or insert a row above the currently selected row. If no row is selected, an insert will add the row to the end.
	Cut Row	Deletes the entire row and holds the contents in the paste buffer.
	Copy Row	Copies the row contents of the currently selected row to the paste buffer.
	Paste Row	Paste contents of the paste buffer to the currently selected row.
	Delete Row	Deletes the currently selected row.
	Move Row Up	Move currently selected row up in the table.
	Move Row Down	Move currently selected row down in the table.
	Resize All Rows	Resize all rows to fit contents.
	Resize All Columns	Resize all columns to fit contents.
	Edit content using Device Explorer	Opens the Device Explorer window.

Use the Edit Target Devices tab on this page to create the device lists (row) you need to define for your test plan. Each list needs to be given a unique name. For each list, add the devices. To add devices to a particular device list, select the list from the table and open the *Device Explorer* window by selecting it from the Edit Target Devices Toolbar.

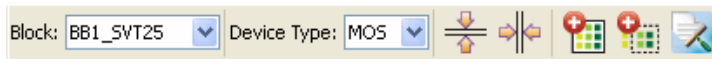
 Use the right mouse button to bring up a popup menu of the same toolbar icon commands.

WaferPro Device Explorer Window



Device Explorer Toolbar

The following figure displays the Device Explorer Toolbar.



The following table describes the various options in Device Explorer Toolbar:

Button	Name	Description
Block: BB1_SVT25	List of available blocks.	
Device Type: MOS	Device Type	List of available device types.
	Resize All Rows	Resize all rows to fit contents.
	Resize All Columns	Resize all columns to fit contents.
	Add All in Block	Adds all devices from the selected block.
	Add Selected	Add only the selected devices to the list.
	Clear	Unselect all selected devices

In the *Device Explorer* Window, use your mouse pointer to select devices from the table to add them to the device list. To select multiple values at one time, either hold down the Shift key to select sequential values or hold down the Ctrl key to select non-sequential values from the table.

After making your selection, use "Add Selected" the toolbar icon or button to add to the device list. If you want to add devices from an entire block, select any device in the

desired block and select the “Add All in block toolbar icon” or Add All button.

When finished, click the **Close** button.

Note
Only one block can be specified per Target Device List. If you try to add another block manually, an error will be issued in the IC-CAP Status window.

The Target Devices Page also has a View List tab which you can use to visualize device list content. Select a Device List using the List Name drop-down list and the viewing area will show all devices selected for that particular device list.

Edit Target Devices View List

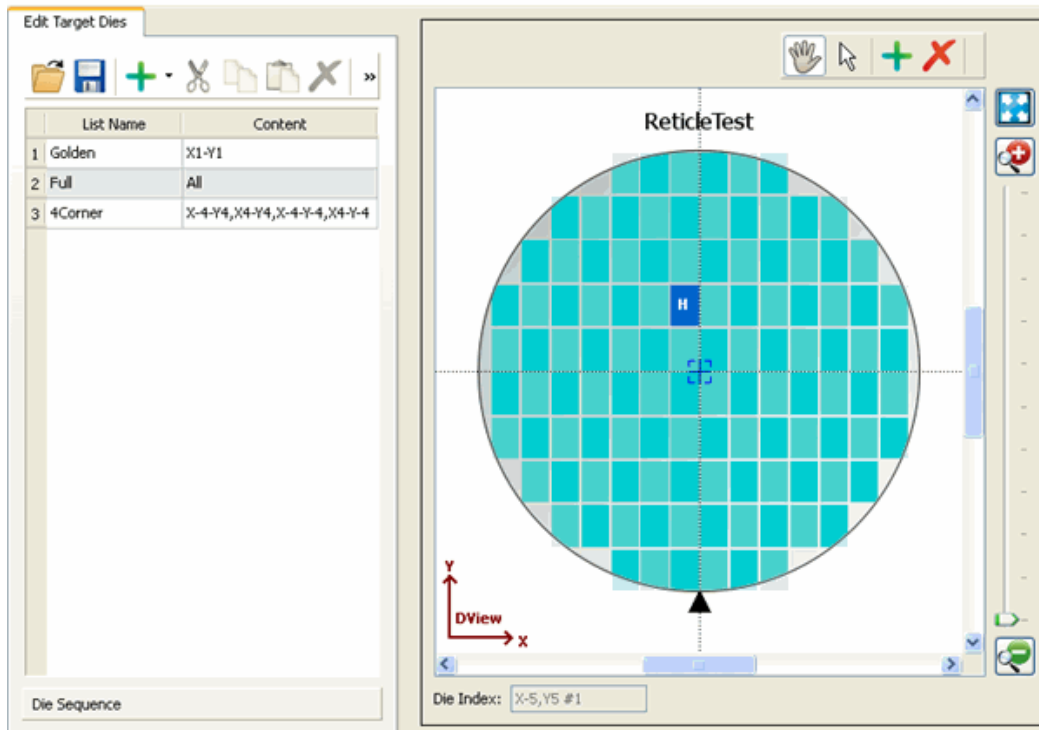
List Name: MOS_25_DC

	Block	Subsite	Dev ID#	Device Name	Polarity
1	BB1_SVT25	MOS01	1	N10/10	N
2	BB1_SVT25	MOS01	2	N10/1	N
14	BB1_SVT25	MOS02	6	N10/0.1	N
15	BB1_SVT25	MOS02	7	N10/0.08	N
16	BB1_SVT25	MOS02	8	N10/0.07	N
17	BB1_SVT25	MOS03	1	N10/10	N
18	BB1_SVT25	MOS03	2	N10/1	N
19	BB1_SVT25	MOS03	3	N10/0.5	N

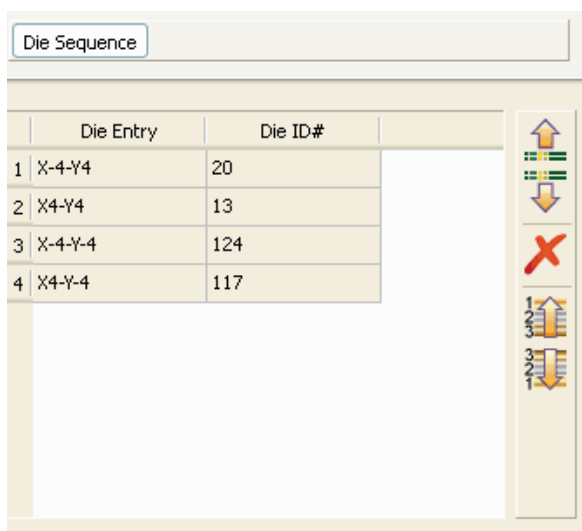
Target Dies

From the Target Dies page, you can specify the test dies to be characterized. You can create multiple Target die lists in the Target Die table. Each Target Die list name must be unique.

You can use the wafer map interface to select the desired die and add them to a Target Die List by selecting the **Add Target Dies** toolbar icon. You can also manually enter your die lists if required.



Press the Die Sequence button to open the Die Sequence pane:



In this pane, you can reset the order of the die measurement sequence using the move up/down toolbar icons or you can sort in ascending or descending die number.

Target Dies Toolbar

The following figure displays the Target Dies Toolbar.



The following table describes the various options in Target Dies Toolbar:

Button	Name	Description
	Load from File	The Target Dies information table can be manually created or automatically created by loading the information directly from a previously saved file.
	Save to File	Save the Target Dies table to a file.
	Add Row	Add a row to the table. Use the pull down menu on this icon to select either to add the row to the end or insert a row above the currently selected row. If no row is selected, an insert will add the row to the end.
	Cut Row	Deletes the entire row and holds the contents in the paste buffer.
	Copy Row	Copies the row contents of the currently selected row to the paste buffer.
	Paste Row	Paste contents of the paste buffer to the currently selected row.
	Delete Row	Deletes the currently selected row.
	Move Row Up	Move currently selected row up in the table.
	Move Row Down	Move currently selected row down in the table.
	Resize All Rows	Resize all rows to fit contents.
	Resize All Columns	Resize all columns to fit contents.

Use the right mouse button to bring up a popup menu of the same toolbar icon commands.

Wafer Map Toolbar

The following figure displays the Wafer Map Toolbar.

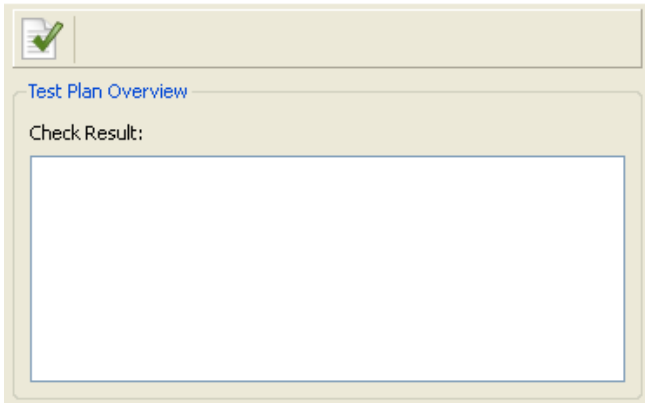


The following table describes the various options Wafer Map Toolbar:

Button	Name	Description
	Hand Tool	Selecting the hand tool, the pointer will turn into a “hand” pointer and allow you to reposition the wafer map inside the viewing area. The hand tool is an alternative to using the scroll bars.
	Selection Tool	Use the selection tool to select die that will you will “deactivate” and remove it from being available when creating the test plan. You can also press and hold the ctrl-key in to select more than one die.
	Add Target Dies	Adds the selected die in the Target Die list.
	Delete Target Dies	Deletes the selected die in the Target Die list

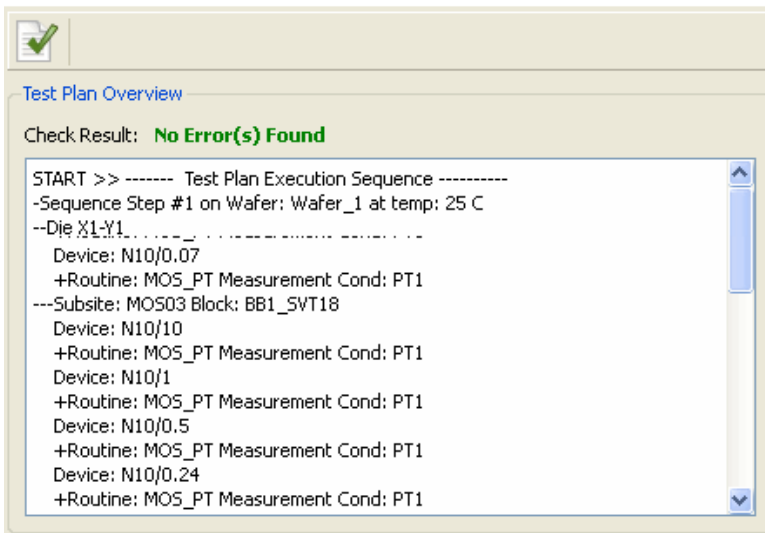
Test Plan Definition

The top level Test Plan Definition page provides a Test Plan Check. You can use this check after defining the complete test plan and WaferPro will perform a validity check on the Test Plan Sequence.



The Test Plan Check will look for errors in specifying Test Die or invalid Measurement Condition names. The results of the validity check will be displayed in the Test Plan Overview area.

The figure below illustrates a validity check on a test plan sequence without errors:



While this figure below shows a validity check on a test plan sequence with an error.

Test Plan Overview

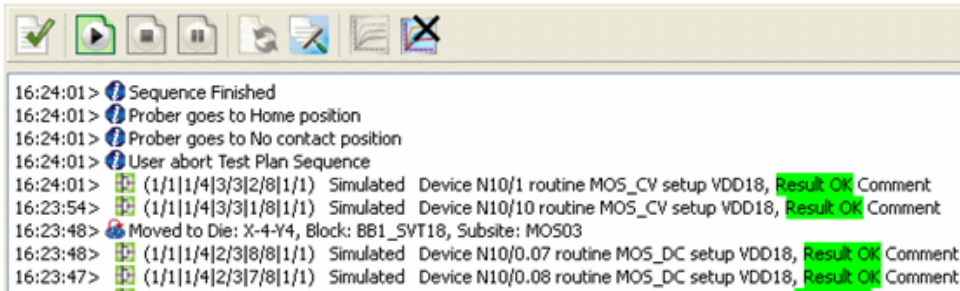
Check Result: **Error(s) Found!**

```
START >> ----- Test Plan Execution Sequence -----
-Sequence Step #1 on Wafer: Wafer_1 at temp: 25 C
--Die X-4-Y4
--Die X4-Y4
--Die X-4-Y-4
--Die X4-Y-4
END >> ----- Test Plan Execution Sequence -----
```

*** Error(s) Reported During Test Sequence Creation:
In Test Plan Sequence, Step 1 Masurement Condition VDD19 is invalid

Test Plan Run

After you set up the Wafer Map and define the Test Plan, the Test Plan Run view allows you to navigate to launch your characterization test plan.



Test Plan Run View

The Test Plan Run view provides an icon tool bar in the work space area to Start, Stop, or Pause test plan execution. It also includes toolbar bar icons to view the test plan, to update the test plan log, to clear the log, and to open or close data displays.

The following figure displays the Test Plan Run View.



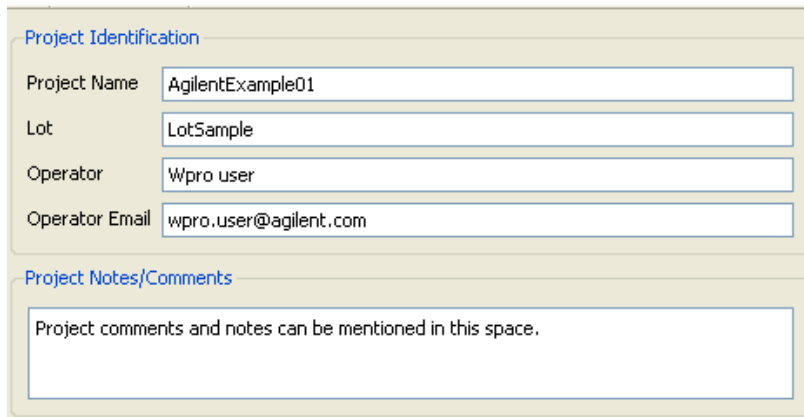
The following table describes the various options Test Plan Run View buttons:

Button	Name	Description
	Show Test Plan	Selecting the Show Test Plan icon will perform a Test plan sequence validity check (same function on the Test Plan node) and display the results of the check in a separate pop up dialog.
	Start/Resume Test Plan Execution	Select to initiate the execution of the test plan or re-initiate it after pausing it.
	Stop Test Plan Execution	Select to abort the test plan execution.
	Pause Test Plan Execution	Select to temporarily pause the currently running test plan. Note: The test plan execution may stop/pause immediately, but will stop/pause upon completion of the current characterization step.
	Update Log	The Update Log Icon only becomes active during test plan execution. The Test Plan execution status log is updated automatically at a time interval specified in the configuration view. Use the Update Log icon to force a manually update.
	Clear Log	During test plan execution, status information is displayed in the test area of the Test Plan Run workspace area. Select this icon to clear this information
	Display Results	Results of completed measurements can be viewed during test plan execution. The Display Results icon will become active by selecting the text for one of the completed measurements displayed in the status log. Once activated, select this icon to open a data display for the results of that measurement. Hint: Use the right mouse button and double click on the completed measurement text as an alternate method to open the display of the desired data.
	Close Displays	Select this icon to close all currently opened data display windows.

Understanding the Project View

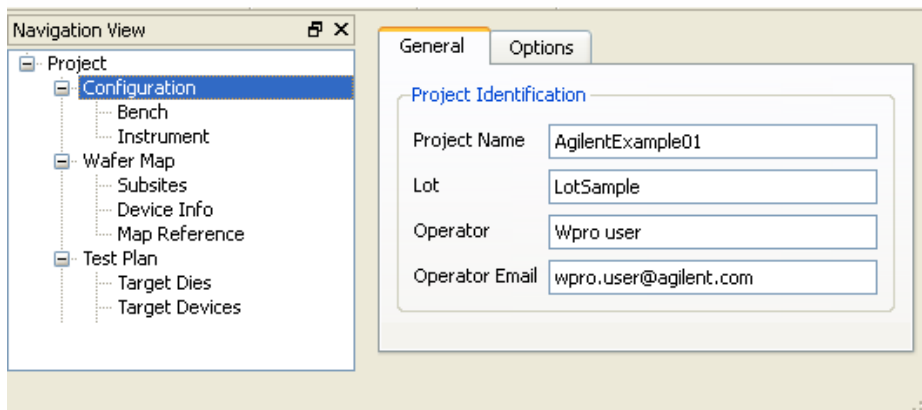
WaferPro consists of workspace windows that are a series of windows with each having its own designated function.

The **Project View** is an area where you can provide information related to your WaferPro project.



The screenshot shows the 'Project Identification' workspace area. It contains two sections: 'Project Identification' and 'Project Notes/Comments'. The 'Project Identification' section has four text input fields: 'Project Name' (AgilentExample01), 'Lot' (LotSample), 'Operator' (Wpro user), and 'Operator Email' (wpro.user@agilent.com). The 'Project Notes/Comments' section has a large text area with the placeholder text 'Project comments and notes can be mentioned in this space.'

The *Project Identification* workspace area can also be accessed from the **General** Tab View in the **Project > Configuration** Node.



The screenshot shows the 'Project Configuration' dialog box. On the left is a 'Navigation View' tree with the following structure: Project (expanded) -> Configuration (selected) -> Bench, Instrument, Wafer Map (expanded) -> Subsites, Device Info, Map Reference, Test Plan (expanded) -> Target Dies, Target Devices. The main area of the dialog has two tabs: 'General' (selected) and 'Options'. The 'General' tab contains the 'Project Identification' section with the same four text input fields as the previous screenshot: 'Project Name' (AgilentExample01), 'Lot' (LotSample), 'Operator' (Wpro user), and 'Operator Email' (wpro.user@agilent.com).

You can set the following project details through the *Project Identification* workspace:

Project Name

Project Name is the text field where you can provide a project name identifier, for example, AgilentExample01.

Lot

Lot is the text field where you can mention the Lot number, for example, LotSample.

 The Lot field value is used as the data storage directory name. This is a required field and should not be left blank or an error will occur during test plan execution.

Operator

Operator is the text field to provide the name of the Operator who will run the test plan, for example, Wpro user.

Operator Email

Operator Email is the text field to enter an Operator's email address, for example, wpro.user@agilent.com.

Project Notes/Comments

You can enter any additional information about the current project.

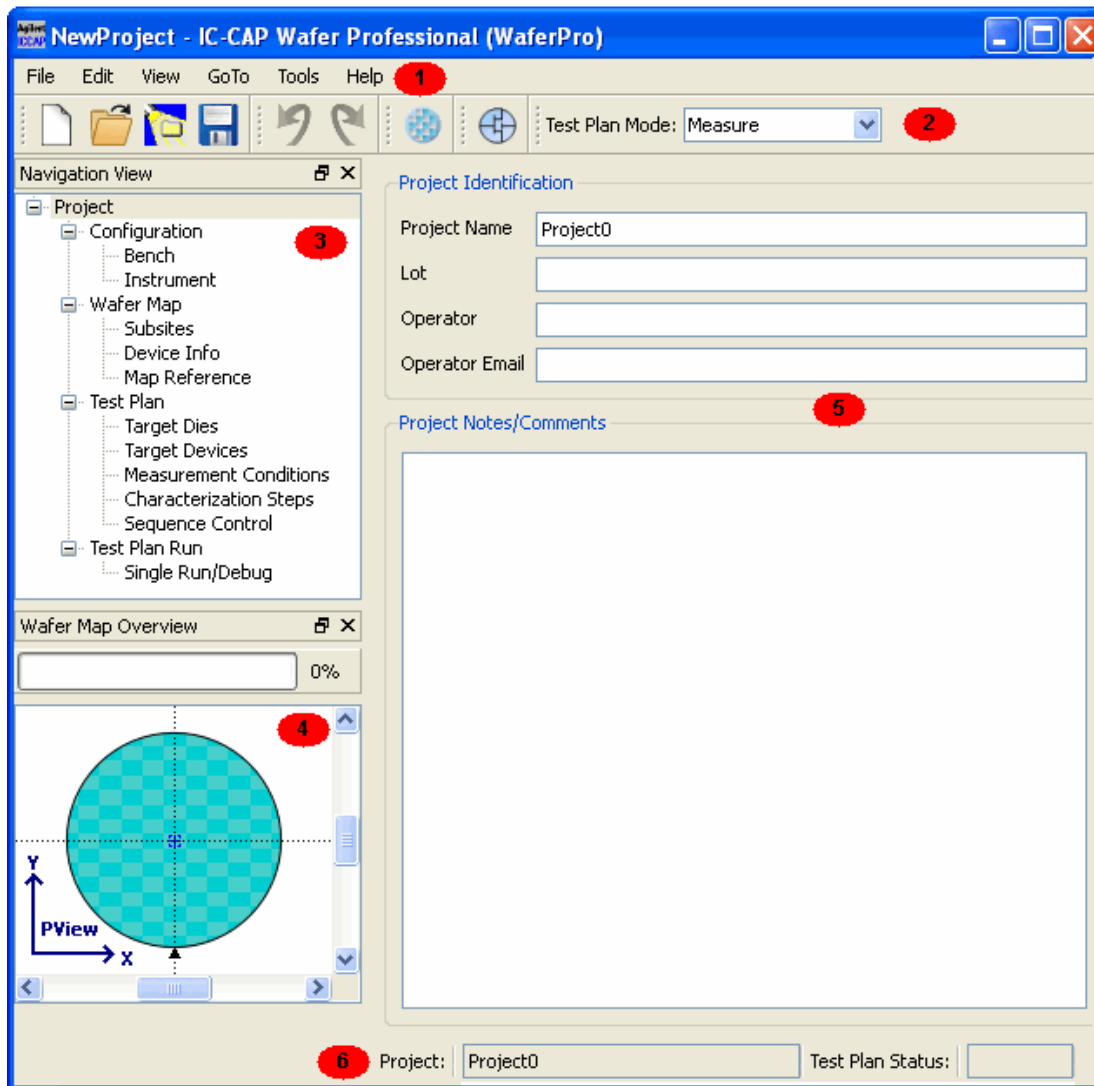
Using the WaferPro Project Interface

Agilent Technologies has introduced a new turn-key DC/CV and RF automated characterization solution to help modeling and device engineers achieving more efficient on-wafer measurements across temperature. This newly designed solution is called IC-CAP IC-CAP Wafer Professional (WaferPro).

The IC-CAP IC-CAP Wafer Professional(WaferPro) runs on the IC-CAP platform as an add on product. The newly designed WaferPro Project Interface allows you to define and manage the WaferPro Test Plan which includes wafer map and device information. It internally links to the core IC-CAP program and utilizes IC-CAP measurement built-in drivers as well as its simulation, programming and graphic engines.

The WaferPro interface comprises several parts that are intended to achieve more efficient on-wafer measurements across temperature.

The following figure highlights the various parts of the WaferPro interface. For information on how to start WaferPro, refer *Starting WaferPro* (waferpro) section.



The WaferPro Project Interface consists of the following:

1

WaferPro Menu Bar (waferpro) - Contains all of the commands used in WaferPro.

2

WaferPro Tool Bar (waferpro) – Contains buttons that are shortcuts for commonly used commands.

3

Navigation View (waferpro) - Displays a hierarchical list of items in a WaferPro project.

4

Wafer Map Overview (waferpro) - Mini view of the main wafer map.

5

Workspace (waferpro) - A context dependent area which will change display according to the project action selected from the Navigation View.

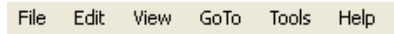
6

Status Bar (waferpro) - Displays the currently loaded project and the status of the test plan execution.

Click on the above links to read more.

WaferPro Menu Bar

The top level WaferPro menus are located on the menu bar at the top of the WaferPro Project Interface.



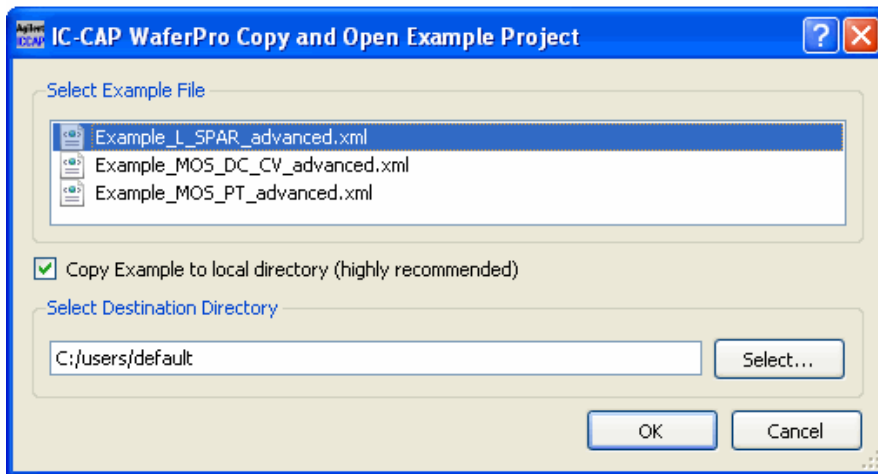
File Menu

The **File** menu includes the following drop-down menus:

- **New Project** - Creates a new WaferPro project.
- **Open Project** - Opens an existing WaferPro project.
- **Open Example Project** - Quickly opens an existing example WaferPro project.

To open an example project:

1. Choose **File > Open Example Project**. The *IC-CAP WaferPro Copy and Open Example Project* window appears.



2. Select an project and click **OK**. For information on the option in *IC-CAP WaferPro Copy and Open Example Project* window, refer *An Example WaferPro Session* (waferpro) section.
- **Save Project** - Saves a new project to a specified measurement directory.
 - **Close Window** - Closes the WaferPro Project Interface.
 - **Exit WaferPro** - Prompts to save the existing open project and exits from WaferPro application.

Edit Menu

The **Edit** menu includes the following drop-down menus:

- **Undo** - The Undo button will "undo" any last action.

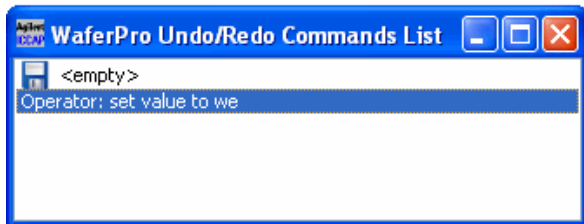
 Not all actions can be "undone". A warning dialog will be issued if a selected action cannot be undone. The command **Tools > Synchronize IC-CAP Routines** is one such example as well as any file load or file read commands.

- **Redo** - The Redo button will repeat any last action that was mistakenly erased.

View Menu

The **view** menu includes the following drop-down menus:

- **Display Undo/Redo Command List** - Displays the list of Undo/Redo Commands in the *WaferPro Undo/Redo Command List* window.



- **Navigation View** - If checked, displays the Navigation View in the WaferPro Project Interface.
- **Wafer Map Overview** - If checked, displays the Wafer Map Overview in the WaferPro Project Interface.

GoTo Menu

The **GoTo** menu includes the following drop-down menus to access the various WaferPro Workspace options:

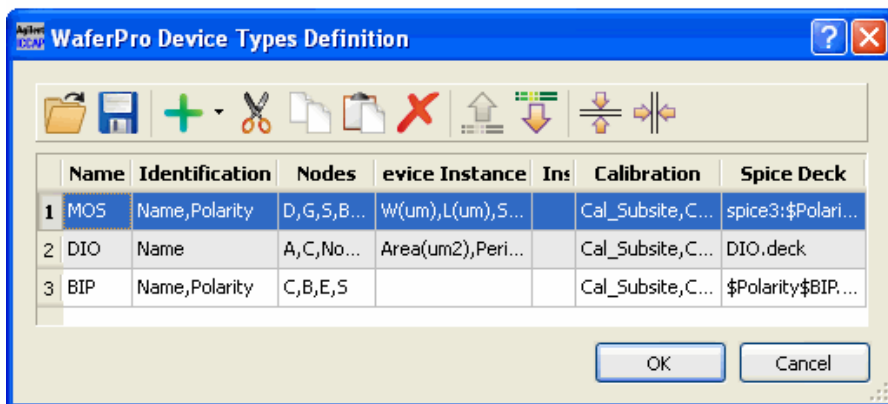
- **Project Configuration and Settings** - Displays the general and optional settings for a project and a test plan, respectively.
- **Bench Settings** - Displays options to set the Wafer Table and bench information.
- **Instrument Settings** - Displays options to set the temperature and instrument control settings.
- **Wafer Map** - Displays the Test Prober View with with Wafer Map description and Die settings.
- **Subsites** - Displays the Subsites information.
- **Device Info** - Displays the Probe Card and Device Type views.
- **Map Reference Settings** - Displays the map reference information.
- **Test Plan** - Displays the test result for the executed test plan.
- **Target Dies** - You can edit the target dies information through Edit Target Dies pane.
- **Target Devices** - You can edit the target devices and view the list of target devices.

- **Measurement Conditions** - Displays the measurement conditions; you can edit the measurement conditions using Edit Measurement Conditions pane.
- **Characterization Steps** - Displays the steps for characterization.
- **Sequence Control** - Displays the sequence control of the test plan.
- **Test Plan Run and Monitor** - Displays the Test Plan Run and Monitor window using which you can view, execute, and pause the test plan.
- **Single Run/Debug** - Displays the mode and settings to run the test plan.

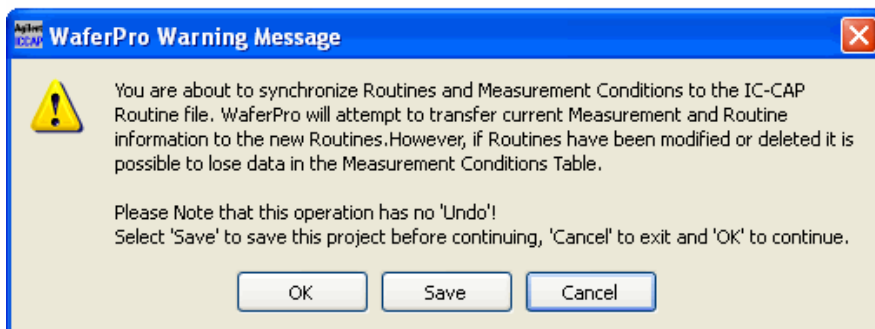
Tools Menu

The **Tools** menu includes the following drop-down menus to perform the device type and WaferPro settings:

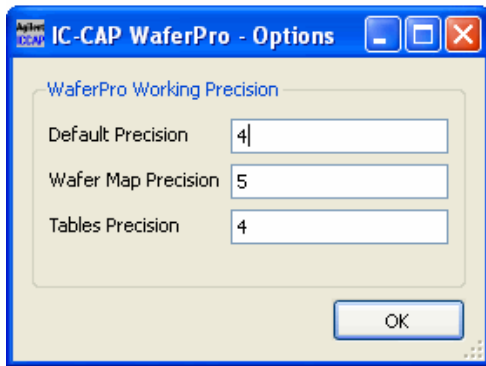
- **Edit Devices Types..** - Displays the *WaferPro Device Type Definition* window which allows you to manage the device type definitions.



- **Synchronize IC-CAP Routines** - If selected, WaferPro will sync the routine and measurement conditions with the IC-CAP Routine file. A *WaferPro Warning Message* appears to display the sync; you can select the appropriate option as per your requirements.



- **WaferPro Options..** - Displays the *IC-CAP WaferPro-Options* window which allows you to set the WaferPro project preferences.

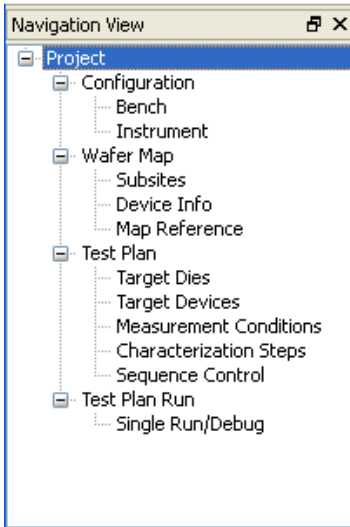


Help Menu

The **Help** menu allows you the access the on-line IC-CAP documentation.

Navigation View

The WaferPro Navigation View provides a hierarchical representation of the WaferPro Project and test plan tasks, as shown in the below figure.



From this hierarchical list of task items, you will navigate through them to configure your project (*Configuration*) (waferpro), specify your wafer floor plan (*Wafer Map*) (waferpro), define your test plan (*Test Plan*) (waferpro), and finally execute your test plan (*Test Plan Run* (waferpro)). Selecting any of the nodes in the Navigation View will update the display on the Workspace area to the right.

The Navigation view can also be minimized or detached from the *Main* window.

To float the Navigation View away from the window

1. With the left mouse button, select the minimize window icon  in the title bar or just double click the title bar.
2. Once detached, to move it, position the pointer in the title bar, press and hold the left mouse, drag to the desired position, and release.

To re-attach Navigation View

1. Position the pointer in the title bar of floating view and press the left mouse button.
2. Drag the toolbar toward the left side of the window and when your pointer is overlapping another view, release.
3. Alternatively, with the pointer in the title bar, you can double click the left mouse button.

To hide Navigation View

1. With the left mouse button, select the close window icon  in the view title bar.

To redisplay the Navigation View

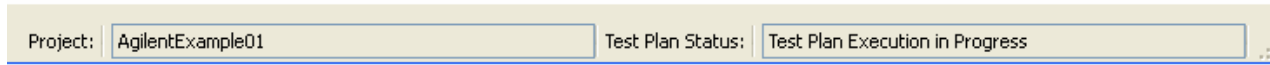
1. Select **View** menu option, then select the view to redisplay.

The Navigation View is organized into the following:

- *Project* (waferpro)
- *Configuration* (waferpro)
- *Wafer Map* (waferpro)
- *Test Plan* (waferpro)
- *Test Plan Run* (waferpro)

Status Bar

The status bar is located at the bottom of the WaferPro window.



It displays the currently loaded project and the status of the test plan execution.

If there is no information on Test Plan Status, the default message is **WaferPro Ready (No execution in Progress)**.

WaferPro Toolbar

The main WaferPro toolbar is shown below:



The toolbars can be repositioned anywhere on the screen. You can move them away from the window and use them like floating palettes or you can dock them along the window's edges.

To float a toolbar away from the window

1. Position the pointer over the dashed separator between icons on the toolbar and press the left mouse button.
2. Drag the toolbar to the desired location and release. When you release the toolbar, the dashed edge rotates to the top of it.

To re-attach a toolbar near the top of the window

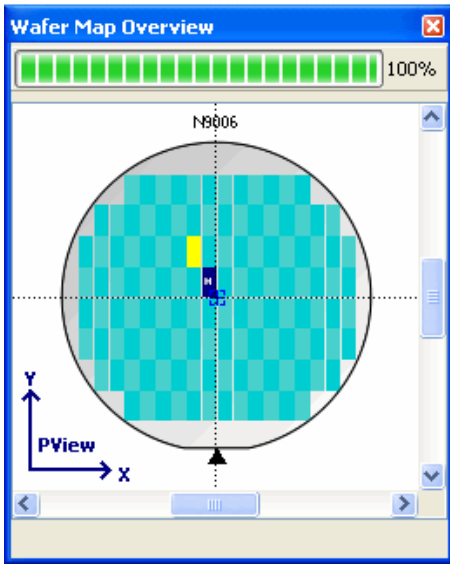
1. Position the pointer in the dashed line at the top of the toolbar and press the left mouse button.
2. Drag the toolbar toward the top of the window and when your pointer is overlapping the menu bar, or another toolbar and release.

The following table describes the various options in WaferPro Toolbar:

Button	Description
	Creates a new WaferPro project.
	Opens an existing WaferPro project.
	Opens an existing example WaferPro project.
	Saves a project to a specified measurement directory.
	Performs Undo operation on the last action.
	Repeats the last action.
	Allows you to change the device type definitions.
Test Plan Mode:	Sets the Test Plan Mode.

Wafer Map Overview

The Wafer Map Overview is a mirror image of the main wafer map (that is, Wafer Map View) as shown in the below figure.



The Wafer Map Overview displays the following information:

- Subsite coordinate information (use the pointer to select)
- Test plan run time messages
- Test plan execution progress

You can minimize or detach Wafer Map Overview from the Main window.

To float the Wafer Map Overview away from the window

1. With the left mouse button, select the minimize window icon  in the title bar or just double click the title bar.
2. To move, position the pointer in the title bar, press and hold the left mouse, drag to the desired position, and release.

To re-attach Wafer Map Overview

1. Position the pointer in the title bar of floating view and press the left mouse button.
2. Drag the toolbar toward the left side of the window and when your pointer is overlapping another view, release.
3. Alternatively, with the pointer in the title bar, you can double click the left mouse button.

To hide Wafer Map Overview

1. With the left mouse button, select the close window icon  in the view title bar.

To redisplay the Wafer Map Overview

1. Select **View** menu option, then select the view to redisplay.

Workspace

A Workspace is a context dependent area which changes the display according to the project task selected from the Navigation View.

Instrument Workspace Area

Instrument

Temperature Control Settings

Long Prober Stabilization (h)

0.000

Short Prober Stabilization (h)

0.000

Wafer Stabilization (min)

0.000

Needle Stabilization (sec)

0.000

Dilatation Method

Prober Driven

Dilatation Coefficient %/deg

1.000

Measure during stabilization

Yes

Stabilize at First Step

No

Instrument Control Settings

Prober

Auto

Manual

Chuck Mode

Auto

Manual

Switch Matrix Control

Auto

Manual

C-V Meter Control

Tis

GPiB

About WaferPro Measurement and Routine Files

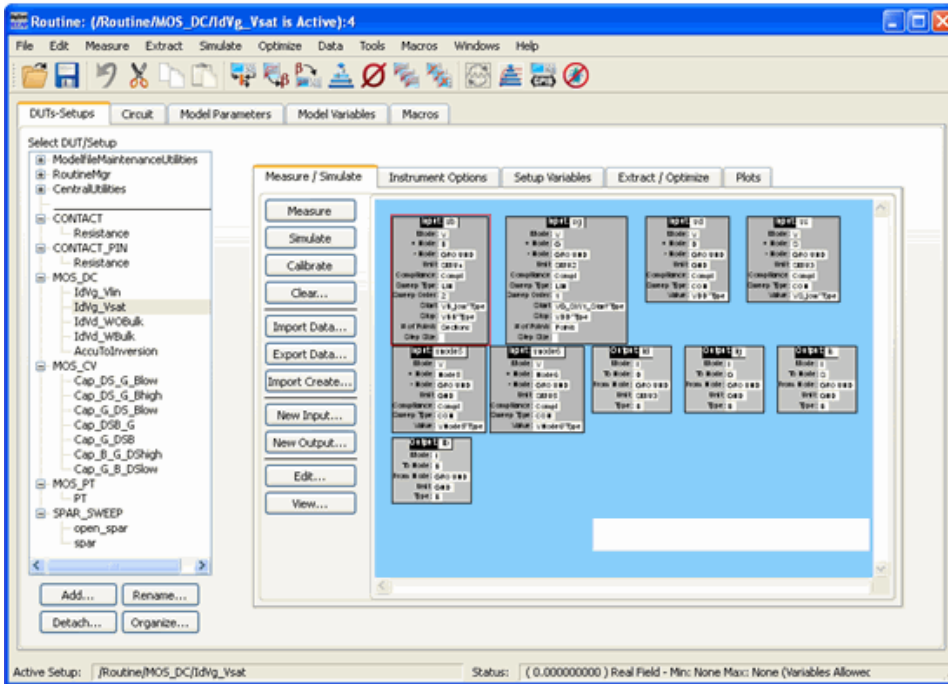
Contents

- *About Measurement Routine File* (waferpro)
- *About WaferPro Routine* (waferpro)
- *Creating a Routine for WaferPro* (waferpro)
- *PEL Functions* (waferpro)
- *Test Plan Execution Flow* (waferpro)

About Measurement Routine File

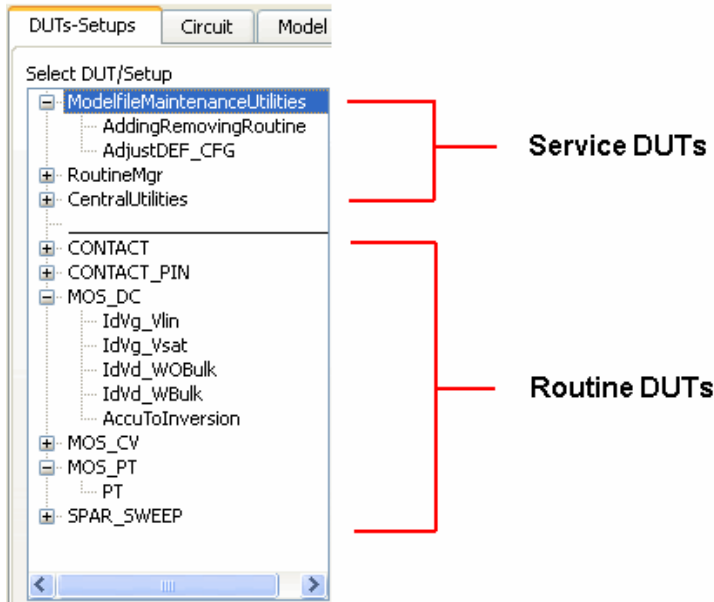
The Measurement Routine Model file contains the library of measurement routines currently provided with WaferPro.

WaferPro provides the flexibility to readily modify one of its current measurement routines or create and add your own routines to the library. These new or updated routines will then be directly accessible for use in a characterization test plan.



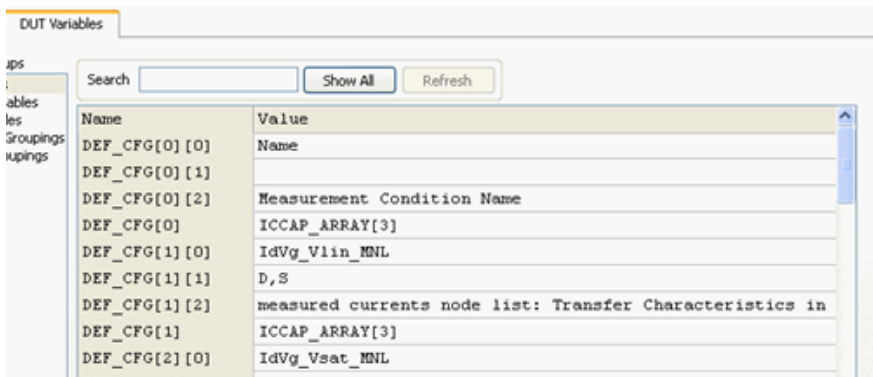
About WaferPro Routine

A WaferPro routine is a set of measurements, configurable with Routine Control Variables (RCV) and managed through user defined algorithms.



Architecturally, a WaferPro Routine is an IC-CAP DUT and utilizes the standard IC-CAP DUT/Setup infrastructure:

- Routines are described with an IC-CAP DUT. DUTs can also be Service DUTs. Service DUTs are used to create/manage Routines and provide other utility functions.
- Each DUT Setup is a Routine measurement (that is, IdVg_Vlin, IdVg_Vsat).
- Routine configuration variables (RCVs) are described in the Routine DUT variable table using the variable array DEF_CFG.

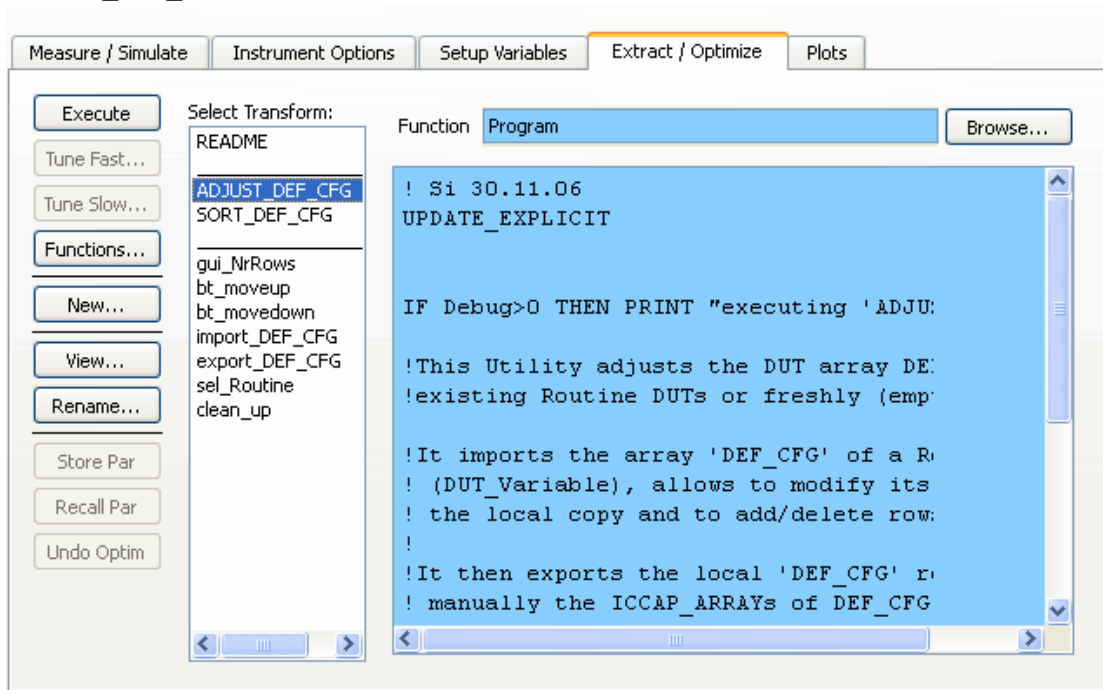


- Routine Setup INPUTs and OUTPUTs are used to define Measurement Conditions.
- Routine Setup plots are used to define routine displayed plots.
- Routine Setup Transforms are used to define measurement and calibration algorithms.

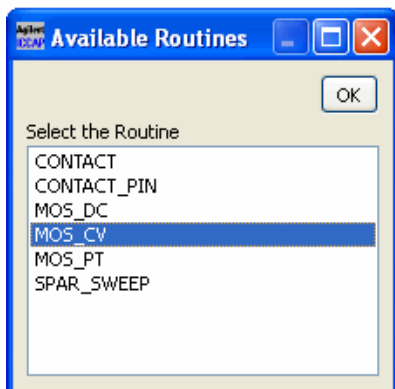
Creating a Routine for WaferPro

To create a routine for WaferPro:

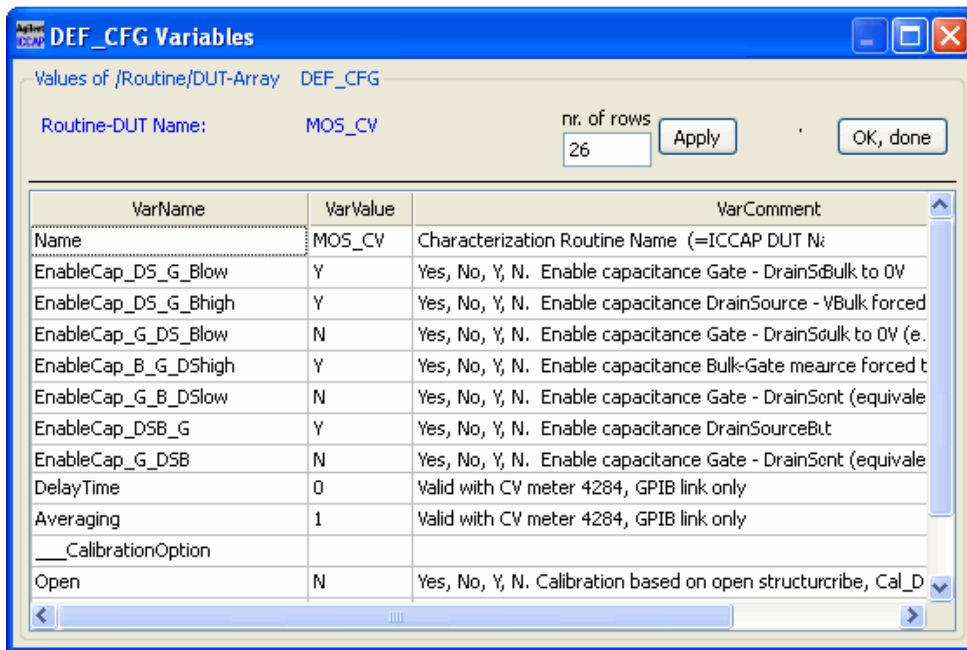
1. Add a new DUT to the routine model file. The name should be the desired Routine name that will be seen in the WaferPro Environment. Alternatively, you can make a copy of an existing Routine and rename it to the desired Routine name.
 2. Create the routine configuration variables table. Use the table to create RCVs to define the measurement conditions that will be selected in the WaferPro environment.
1. Run macro: /Routine/ModelMaintenanceUtilities/AdjustDEF_CFG/
ADJUST_DEF_CFG.



2. In the dialog, select the routine you are working on and click **OK**.



3. In the *DEF_CFG Variables* window, define the variables you want to create in "nr. of rows", then click **Apply**.



4. Complete the table, providing for each line: The variable name, default value and associated comment.
5. Click **Done**. Each variable created in the previous table are now accessible in your routine/targets macros and/or inputs and outputs fields.
6. If you need to reorganize or modify the DEF_CFG table, you can use macro: /Routine/ModelMaintenanceUtilities/AdjustDEF_CFG/ SORT_DEF_CFG.

i

- The first variable in the DEF_CFG Table must be "Name" to identify the characterization Measurement Condition.
- In a routine, variable names must be unique.
- If the measured node list (MNL) of a setup is configurable, then a variable named SetupName_MNL must be created. WaferPro will automatically enable/disable measurement units corresponding to the specified nodes in this variable.
- Variables used for measurement precision definition must have names with the suffix "_Precision" added to it.

3. Add as many Setups as desired to your routine (DUT).
4. For each Setup, define the INPUTS and OUTPUTS.
5. Define Setup Transforms as needed for measure or calibration purpose.

Each Setup MUST include the following two PEL transforms:

INIT_SETUP

It enable/disable setups depending on the defined Measurement Conditions. This transform should:

- Reset INPUTS and OUTPUTS to their initial configuration.
- Enable/Disable target execution depending on setup variables values.
- Load specified Precision.iot instrument option file.
- Check / modify INPUT and OUTPUT configuration. For example round bias sweep step to the required resolution (use /Routine/CentralUtilities/CheckInstrResol/CheckInstrResolSetup macro), or set CV meter frequency and oscillation level.

MEASURE_SETUP

Used to support "Measure", "Simulation" and "Load From Record" modes depending on the Test Plan execution mode selected in the WaferPro environment.

- Depending on specified Session/Bench, run measure, simulation or load mdm data file.
- If INPUTS or OUTPUTS are applied to shorted nodes, a Routine test circuit to emulate switching matrix role in SPICE simulation is needed. To do this, use the macro provided in the centralUtilities Service DUT:
/Routine/centralUtilities/SimulData/Emul_Conn.
- Run mdm data storage if desired. To do this, use the macro provided in the centralUtilities Service DUT:
/Routine/centralUtilities/SaveDataRoutine/SaveDataRoutine.

An optional 3rd transform, POSTPROCESS_SETUP to support "Post Processing" mode may be specified. Use this transform to define:

- Data calibration algorithms.
- Parametric Test extraction and storage algorithm.
- Data validity control algorithms.

PEL Functions

WPro_lookup_info() and WPro_set_info() Functions

When implementing new measurement routines, two PEL functions, `WPro_lookup_info()` and `WPro_set_info()`, are available to get test plan related information entered in the WaferPro project by the user or set test plan default related information in the WaferPro project.

```
WPro_set_info (arg1, arg2, arg3)
```

```
WPro_lookup_info (arg1, arg2, arg3)
```

Where,

`arg1` is a string used to specify the type of information to get/set.

`arg2` is a string used to specify, depending on the selected type, the exact information to get/set.

`arg3` is a string used to specify:

- The value that will be affected by the object defined with `arg1` and `arg2` when used with `WPro_set_info()`.
- the name of the IC-CAP setup variable or model variable that will be updated with the value of the object defined with `arg1` and `arg2` when used with `WPro_lookup_info()`.

Return is Always 1.

Examples

```
WPro_lookup_info()
```

```
!get .iot file name
```

```
bRes = WPro_lookup_info("Precision", VAL$(IdVg_Vsat_Precision), "IotFileName")
```

```
WPro_set_info()
```

```
!set the mdm file name to save the data
```

```
dummy = WPro_set_info("Run", "SaveMdm", VAL$(FileName))
```

WPro_lookup_info() Table

Type (arg1)	Variable(arg2)	Values
Run	Mode	[Simulation, Measure, Load Form Record, PostProcess]. Get data origin: Spice simulation, on silicon measure or mdm & csv files. Note: when running a Debug/Single Routine, the returning mode is the one specified in the Debug/Single Page.
	Display	[TRUE, FALSE]. TRUE if Session/Plot Display is enabled.
	Stop	[TRUE, FALSE]. TRUE if the "Stop" button from Test Plan Execution View has been pressed; otherwise FALSE.
	StoragePolicy	[NoRecord, MDM_Record, DB_Record, MDM_DB_Record]. Get value of session/Storage Policy option. Note: DB_XXX values are not

IC-CAP 2011.01 - IC-CAP Wafer Professional (WaferPro)
currently available.

	SaveMDM	[Mdm full file name, Empty]. Get/Set mdm file full name for current device, target measured data storage. Empty value if function fails (invalid storage path). Note: Used by data saving PEL routine to retrieve the MDM file name to be used (get). It will try to create a full path if it does not exist already. After successful saving to MDM, used by data saving PEL to tell WaferPro that the file was saved. WaferPro will add the file to MDM .csv file for the current device.
	LoadMDM	[Mdm full file name, Empty]. Get the mdm file full name for current device, target data loading. Empty value if function fails (invalid storage path). Used by MEASURE_SETUP when mode is "Load From Record" to retrieve the file where MDM data are stored. This will provide the name of the MDM file but unlike SaveMDM, it will not attempt to create a path.
Context	SetupList	[ICCAP_ARRAY of current routine Setups]. Get the list of current routine SETUP names.
	Date	Get current date.
	Bench	Get session/bench name.
	Temperature	Get / Set current temperature.
	Lot	Get / Set current lot name.
	Operator	Get / Set current operator name.
	Reticle	Get / Set current reticle name.
	Wafer	Get / Set current loaded wafer name.
	Die	Get / Set current die under test name.
	DieList	Get / Set currently used die list name.
	Block	Get / Set the current measured block. If Set, then current scribe and device need to be specified.
	BlockOption	Get the current block option.
	Subsite	Get / Set current measured Subsite. If Set, then current device needs to be specified.
	Routine	Get / Set current routine. If Set, then current setup needs to be specified.
	MeasCond	Get / Set current Measurement Condition
	Device	Get / Set current device in subsite ID.
	DeviceName	Get / Set current device name.
	Setup	Get / Set current SETUP name.
Map	DieX	Get Die X size (mm), in WaferPro referential (based on Wafermap documentation view and reference)
	DieY	Get Die Y size (mm), in WaferPro referential. (based on Wafermap documentation view and reference)
	IndMovX	Get Die X size (mm), in prober referential (based on documentation, wafer angle and dilatation coefficient)
	IndMovY	Get Die X size (mm), in prober referential (based on documentation, wafer angle and dilatation coefficient)
Subsite	Name	Get current subsite name.
	Pos.X	Get current subsite X position in um, as defined in the subsite definition table
	Pos.Y	Get current subsite Y position in um, as defined in the subsite definition table
Device	Nodes	[ICCAP_ARRAY of device nodes] Get the list of current device nodes.
	DeviceInstances	[ICCAP_ARRAY of spice instances names] Get the list of current device spice instances parameters names.
	TechnoInstances	[ICCAP_ARRAY of techno instances names] Get the list of current device techno instances parameters names.

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	Calibration	[ICCAP_ARRAY of calibration instances names] Get the list of current device calibration instances parameters names.
	SpiceDeck	Get the full name of the SPICE deck file to use for current device simulation.
	MatrixDeck	Get the full name of the Switch Matrix deck file to use to emulate Switching matrix when simulating current device with current routine/setup.
	PTList	[ICCAP_ARRAY of Parametric test names] Get the list of current device parametric test stored by the current routine.
	Comment	Get / Set comments about the current routine execution on the current device.
	Status	[OK, NOK, Warning] Get/ Set current routine on current device execution status
	\$AnyInstName	Get / Set current device instance parameter \$AnyInstName value.
	\$AnyPTName	Get / Set current device parametric test \$AnyPTName result.
MeasCond	VariableList	[ICCAP_ARRAY of setup variables names] Get the list of current Measurement Condition variables names.
	\$MeasCondVarName	Get current Measurement Condition variable \$MeasCondVarName value.
Hardware	\$InstrUnitName	Get pin(s) connected to the instrument unit \$InstrUnitName.
	PIN[\$NodeName]	Get pin(s) connected to the device node \$NodeName
	\$InstrType	Get driver name of the instrument of type \$InstrType. \$InstrType supported values are defined in WproBenchList.cfg; typical \$InstrType are: Prober, Matrix, Chuck.
	\$InstrType[\$InstrVar]	Get Instrument of type \$InstrType variable \$InstrVar value.
Precision	\$Precision	For the specified \$Precision variable value , get the corresponding iot file full name.

WPro_set_info() Table

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Type (arg1)	Variable(arg2)	Values
Run	SaveMDM	[Mdm full file name, Empty]. Set mdm file full name for current device, target measured data storage. Empty value if function fails (invalid storage path). Note: Used by data saving PEL routine to retrieve the MDM file name to be used (get) It will try to create a full path if it does not exist already. After successful saving to MDM, used by data saving PEL to tell WaferPro that the file was saved. WaferPro will add the file to MDM .csv file for the current device.
	Setup	[TRUE, FALSE] Enable/Disable the current SETUP's measure. Note: Used in INIT_SETUP to enable/disable the current SETUP in the running Routine (usually based on a flag in the Measurement Condition).
Context	Temperature	Get / Set current temperature.
	Lot	Get / Set current lot name.
	Operator	Get / Set current operator name.
	Reticle	Get / Set current reticle name.
	Wafer	Get / Set current loaded wafer name.
	Die	Get / Set current die under test name.
	DieList	Get / Set currently used die list name.
	Block	Get / Set the current measured block. If Set, then current scribe and device need to be specified.
	Subsite	Get / Set current measured Subsite. If Set, then current device needs to be specified.
	Routine	Get / Set current routine. If Set, then current setup needs to be specified.
	MeasCond	Get / Set current Measurement Condition.
	Device	Get / Set current device in subsite ID.
	DeviceName	Get / Set current device name.
	Setup	Get / Set current SETUP name.
Device	Comment	Get / Set comments about the current routine execution on the current device.
	Status	[OK, NOK, Warning] Get/ Set current routine on current device execution status.
	\$AnyInstName	Get / Set current device instance parameter \$AnyInstName value.
	\$AnyPTName	Get / Set current device parametric test \$AnyPTName result.

Test Plan Execution Flow

This section describes the Wafer Pro Test Plan Execution Flow. When developing your own Routines for Wafer Pro, you should have a good understanding of this flow in order to ensure that your new routines provide all the elements needed for Wafer Pro to execute correctly with them.

When running a Testplan, WaferPro executes a sequence of device measurements at specific locations across the wafer(s) as defined in the current project. The sequence of wafer, die and sub-site locations, along with the measurement applied to each device can be viewed in the WaferPro window when selecting the **Check Test Plan** icon in the **Test Plan** section tab.

During the Testplan execution, WaferPro will load a wafer and set its temperature according to the Testplan sequence table. At each step, the WaferPro engine will step through each sub-site of the selected die list. When WaferPro steps the prober to a specific sub-site, it first connects one of the devices using the Switch Matrix driver (according to the pin configuration specified in the Device Info table). Based on whether the device belongs to one or more Target Device List and Characterization Steps, WaferPro executes every Routine/Measurement Condition specified in the test plan for the target device.

Once the device is connected, to enable a device measurement with a specified Routine and Measurement Condition, WaferPro testplan engine will execute the following sequence:

- Reset all the measurement data in the Routine DUT (all the setups).

 When designing new Routines and their setups, make sure PEL transforms in the setups are not giving floating point errors when measured data in the setup are zero-ed.

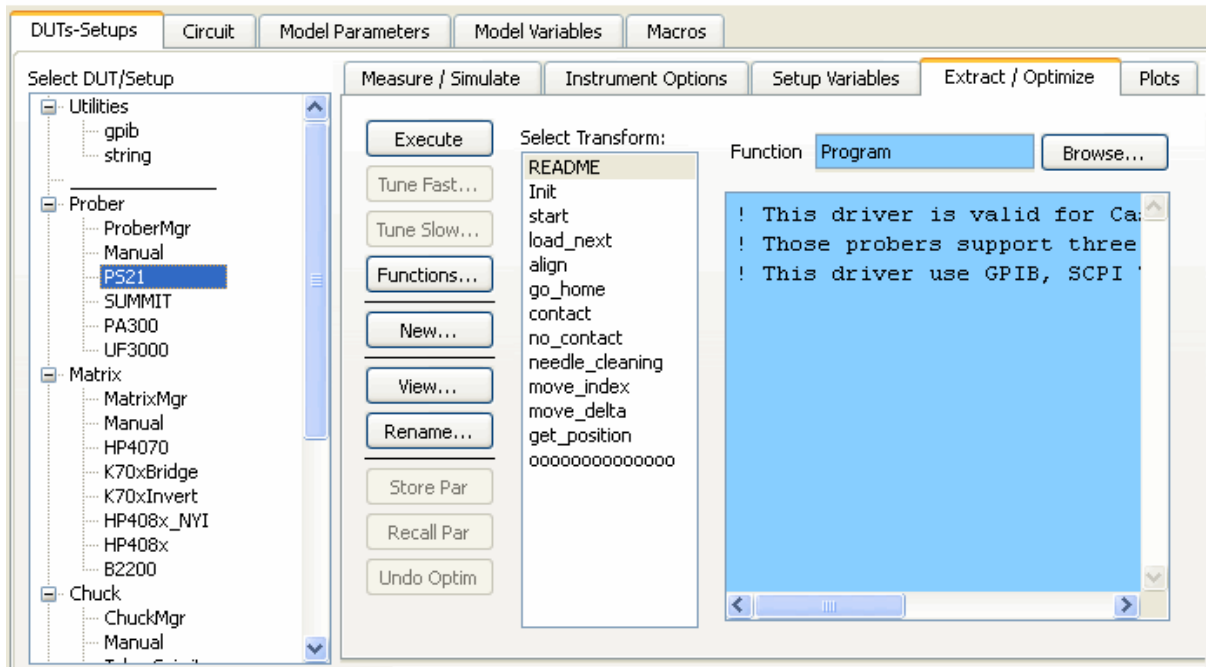
- Set the Context Routine Model Variable (e.g. Wafer, Die, DieList, Setup, Routine, MeasCond, Device, etc.). These variables are at the IC-CAP Model Variable level.
- Set the Routine Control Variables to Measurement Condition values. These are the Routine DUT variables (for example, VG_START, VG_STOP, etc.)
- For each Setup in the Routine DUT, call the INIT_SETUP transform (see INIT_SETUP doc...). The initialization is executed using the same sequence the setups are ordered within the DUT.

 Each Setup has the ability to enable/disable itself (typically this is based on the Measurement Condition Variables) during the execution of the INIT_SETUP transform.

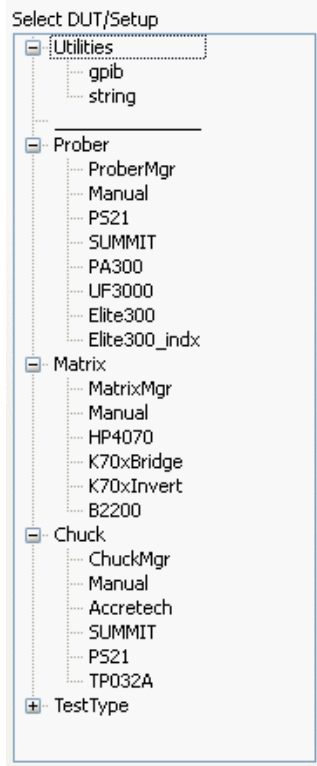
- For each enabled setup, call either MEASURE_SETUP ("Measure", "Simulation", "Load From Record" Test Plan modes) or POSTPROCESS_SETUP ("PostProcess" Test Plan mode). These transforms execute the actual Measurement or Simulation or Data Access (depending on the mode) and save the data to MDM files.
- Before switching to the next device in the sub-site (if any), WaferPro will first ground all the pins and then make a new device connection.

About WaferPro Driver File

The Driver model file includes the library of PEL drivers for the auxiliary instrumentation (Prober, Chuck, and Switching matrix) currently supported by WaferPro.



The model file organizes the drivers for each instrument type into a DUT and each Setup in the DUT includes the PEL driver functions for a specific instrument.



Each instrument type DUT also includes a "Instrument Type manager" Setup which provides the functional interface between the WaferPro environment and the driver, that is, ProberMgr.

 The PEL functions in the “manager” setup should not be modified or Wafer Pro execution may not work correctly

New drivers however can be easily added to the library by implementing a few simple PEL functions that will be called by WaferPro during test plan execution. The Driver model file also includes a Utilities DUT which provide some gpib and string manipulation functions to assist in developing a new driver.

The following tables list the driver PEL function requirements for each instrument type. To create your own auxiliary instrument driver, it is recommended to copy one of the existing driver Setups and modify the functions as required for your specific instrument.

Prober Driver

Macro	Purpose
Init	Get instrument GPIB interface, address and command suffix.
Start	Switch prober to GPIB controlled mode.
Go_home	Move prober back to its home position.
Contact	Contact wafer / probe card.
No_contact	Un-contact wafer / prober card.
Move_delta	Move prober of delta_x delta_y μm relative to current position.
Align	Run an automated wafer and probe card realignment.
Needle_cleaning	Perform a needle cleaning without changing current prober position.
Load_next	Load and preload wafers from specified slots.
Move_index	Move prober of delta_x delta_y die relative to current position.
Get_position	Update variables current_y and current_x with prober position in μm .

Matrixes Driver

Macro	Purpose
Init	Get instrument GPIB interface, address and command suffix.
Connect	Connect specified unit to specified pin.
Disconnect_all	Disconnect all pins.
Ground_all	Ground all pins.

Thermal chucks Driver

Macro	Purpose
Init	Get instrument GPIB interface, address and command suffix.
Get_temperature	Update curr_temperature variable with chuck current temperature in Celsius.
Set_temperature	Set chuck to specified temperature in Celsius.

Working with Configuration and Output Files

Configurations Files

WaferPro uses a number of configuration files. These files contain definitions to configure test benches, define Instrument precision options, declare Device Types, probe card dependent pad to pin relation, or spice decks for simulations.

Bench Description File

The `wproBenchList.cfg` file is used to define bench descriptions and is located in the `$ICCAP_WPRO_ROOT/config` directory. You can open this file in any simple text editor.

 Do not use any word processing software that will embed additional formatting characters.

This file is used to define the test bench configurations that you will use with WaferPro.

The following is a template of the Bench definition. The fields denoted in square brackets `[]` are the fields that have to be set for the bench being created.

Bench Definition Template

```
1 : <SECTION Title="Bench[#]">
2 : <VAR Name = "[BenchName]"></VAR>
3 : <VAR SlotCount = "[#]"></VAR>
4 : <VAR SMUCount = "[#]"></VAR>
5 : <VAR CVTisSupport = "[Boolean]"></VAR>
6 : <VAR ProberCoordinates = "[ProberCoordinates]"></VAR>
7 : <VAR GoHomeBeforeDieMove = "[Boolean]"></VAR>
8 : <TABLE Title="InstrumentPilot" Format="csv">
9 :
"Type","Name","Controller","Address","CmdSuffix","PinsCount","GROUND","CMHigh","CMLow","SMU1","SMU2",
",","SMU3","SMU4"
10 :
"Prober","[ProberName]","[InterfaceName]","[GpibAddress]","[TerminationCharacter]","", "", "", "", "",
",","", "", ""
11 : "Scanner","[ScannerName]"," [InterfaceName]"," [GpibAddress]","
[TerminationCharacter]","7","9","7","8","1","2","3","4","5"
12 : "Chuck","[TempControllerName]"," [InterfaceName]"," [GpibAddress]","
[TerminationCharacter]","", "", "", "", "", "", "", "", ""
13 : </TABLE>
14 : </SECTION>
```

In the above code,

- Line 1: Benches are numbered 1 to N. For a new bench, increment the bench number by 1 from the last bench number used.
- Line 2: Provide a descriptive bench name. The name should be identical to the file name given to the hardware configuration file for the bench.
- Line 3: If the test bench will support cassette loading, set this to the cassette slot count; otherwise set it to 1.
- Line 4: Set # to the number of SMU's your parametric analyzer is configured for.
- Line 5: Set to TRUE if the CV meter can be driven through the TIS interface (407x/8x

systems).

- Line 6: Set the X-Y coordinate direction mapping, ie X-Right, Y-Up.
- Line 7: Set to TRUE if prober will return to HOME position before every move; else FALSE.
- Line 8: No modifications.
- Line 9-12: The instrument pilot table is used to define configuration of context control instruments (prober, chuck, switching matrix). For each table column, the corresponding instrument value is defined.
 - Type: Describes the instrument family. In Wafer Pro Driver model file, it corresponds to the instrument driver DUT (Refer to Link to Driver Model file doc).
 - Name: Is the instrument driver name.
 - Controller: If exists, name of the computer GPIB controller.
 - Address: If exists, Instrument GPIB address.
 - CmdSuffix: GPIB command terminator character (i.e, \r or \n).

The fields after `CmdSuffix` are variable in number and are meaningful for a switching matrix only. They are used to define the measurement instruments units to switching matrix row connections.

Instrument Precision Files

The Instrument Precision files used with Wafer Pro are standard IC-CAP instrument option files (.iot) that are used to define a desired characterization precision level for the supported DC measurement instruments and to define any other desired instrument options for. These files are located in `$ICCAP_WPRO_ROOT/config` directory.

 It is recommended to use IC-CAP to create your Instrument Precision Files, but if you manually edit the file, do not use any word processing software that will embed additional formatting characters.

The example below illustrates an Instrument Precision File created for a bench that contains an HP4156 Semiconductor Parametric Analyzer and an Agilent E4980 Precision LCR Meter. Note the integration time (Integ Time) for the HP4156 has been set to "S" for Short Integration (fast measurement, but less accuracy). Also note in the example below the use of variables for some of the instrument options for the Agilent E4980. These variables are declared in the PEL function `/Routine/RoutineMgr/MeasureRoutine/InitRoutine` which is a transform included in the Wafer Pro Routine model file and then exposed to the Wafer Pro environment by adding Routine Control Variables (RCVs) to a specific measurement routine (See the RCV's defined in the DUT Variable table of the MOS_DC routine).

Example

```
data
{
TABLE "AgilentE4980.7.17"
{
View par_var_table 1 "V530 C18 C15"
element 0 "Hold Time" " HOLD "
element 0 "Delay Time" " DELAY "
element 0 "Meas Freq" " CV_FM"
element 0 "Integ Time" "S"
element 0 "Osc Level" " CV_SIG"
element 0 "Osc Mode [V,I]" "V"
```

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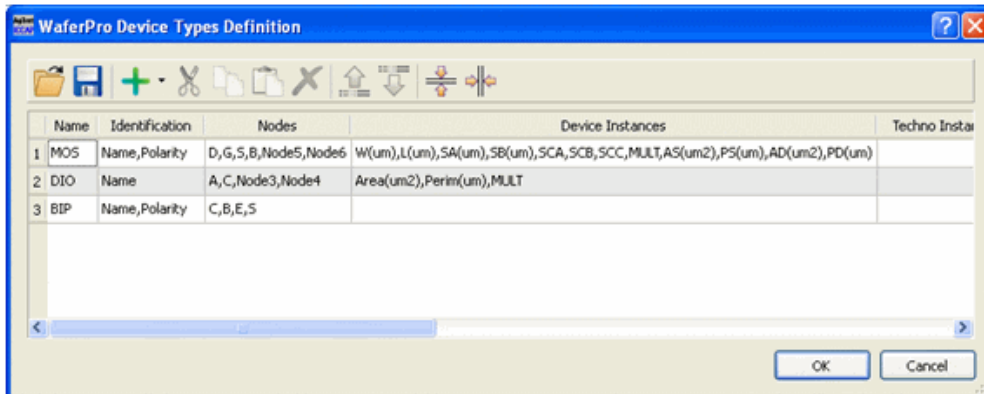
```

element 0 "Averaging [1-256|1-256]" " AVERAGE "
element 0 "Cable Length" "1"
element 0 "Delay for Timeouts" " 0.000 "
element 0 "Init Command" "CORR:OPEN:STAT OFF;:CORR:LENG 1"
}
TABLE "HP4156.4000.24"
{
View par_var_table 0 "V530 C18 C15"
element 0 "Use User Sweep" "No"
element 0 "Hold Time" " HOLD "
element 0 "Delay Time" " DELAY "
element 0 "Integ Time" "S"
element 0 "Power Compliance" " 0.000 "
element 0 "Pulse Unit" ""
element 0 "Pulse Base" " 0.000 "
element 0 "Pulse Width" " 1.000m"
element 0 "Pulse Period" " 10.00m"
element 0 "Delay for timeouts" " 1000.0 "
element 0 "Init Command" ""
}
}

```

Device Type Definitions File

The `wproPredefinedTypes.csv` file is the default Device Type definition file used with Wafer Pro. This file is located in `$ICCAP_WPRO_ROOT/devices` directory and provide Device definitions for Additional Device Types can be added from within the WaferPro environment by selecting **Tools > Edit Device Types...** (or corresponding Toolbar Icon) from the main WaferPro Project window.



Spice Deck Files

The Spice Deck files are used to define spice models that will be used to simulate devices when Wafer Pro is used in simulation mode and are located in `$ICCAP_WPRO_ROOT/simulation` directory. Spice decks are ASCII text files that need to satisfy IC-CAP circuit description syntax.

Example: NMOS.deck

```

.OPTIONS gmin=1e-18
.SUBCKT MOSTEST 1=D 2=G 3=S 4=B 5=Node5 6=Node6
RW 5 0 1G
RDW 6 0 1G
\#echo M1 1 2 3 4 MMOD W=$mpar(W=0.4u) L=$mpar(L=0.28u)
.MODEL MMOD NMOS
+ LEVEL = 8 VERSION = 3.2.2 TNOM = 27

```

```
+ MOBMOD = 1 CAPMOD = 3 NOIMOD= 1 PARAMCHK= 1
+ DELTA = 0.01 TNOM = 27 TOX = 7.5E-9
+ NCH = 1.7e17 XJ = 1.5E-7 VTH0 = 0.69 K1 = 0.53
+ K2 = \-0.013 K3 = 0 K3B = 0 DSUB = 0.56
+ U0 = 675 UA = 2.25E-9
.ENDS
```

Probe Card Pad to Pin Description Files

The Probe Card Pad to Pin Description Files specify the relationship between the probe card plot identifier (that is, device pad) to the corresponding switching matrix relay number (that is, probe card pin). These files are simple ASCII text files and are located in *\$ICCAP_WPRO_ROOT/instr* directory.

Example

```
Plot Relay
\-----\
1 22
\-----\
2 26
\-----\
3 20
\-----\
4 28
\-----\
5 18
\-----\
6 30
\-----\
7 16
\-----\
8 32
\-----\
9 14
\-----\
10 34
```

 Do not use any word processing software that will embed additional formatting characters.

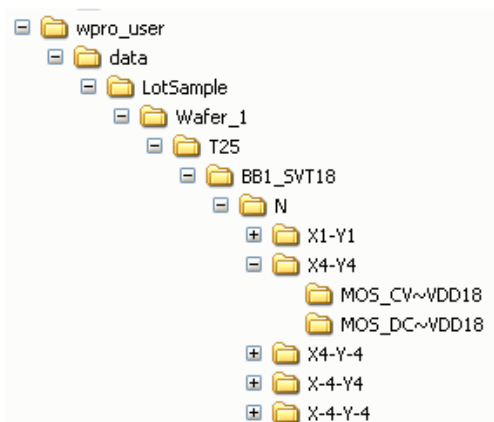
Output Files

Data Files

When a characterization test plan is run, WaferPro produces a large amount of characterization data files. These data files are IC-CAP .mdm file format and are stored in a directory hierarchy created under the user specified Storage Path directory set from the WaferPro Configuration View as described below:

- Under the Storage Path directory a lot directory is created then directories for each measured wafer. The Lot directory name used is from the Lot field specified from the WaferPro Configuration page.
- Under each wafer directory are created directories for each measured temperature specified.
- Under each temperature directory are created directories for each measured block.
- Under each block directory are created directories for each Polarity.
- Under each polarity directory are created directories for each measured Die.
- Under each die directory are created directories for each Routine Measurement Condition. Both the Routine Name and Measurement Condition will be used separated by a "~".

- Finally, in Routine Measurement Condition directories the MDM data files are stored.



The naming convention used for the MDM files is to use the following elements project separated with a tilde "~" between each:

```
<SubsiteName>~<DeviceName>~<DeviceID>~<RoutineSetupName>.mdm
```

Where,

SubsiteName is the subsite name specified in the Subsites table.

DeviceName is the device name defined in the Device Info device definition table.

Device ID is the the row number of the device listed in the "Device definition" table.

RoutineSetupName is the name of the Routine measurement setup used to create the data.

 If a "/" is used in the Device Name, it is replaced by "~" in the MDM file name.

Example

```
MOS01~N10~0.5~3~IdVd_WOBulk.mdm
```

Characterization Report File

A Characterization Report file is generated when the test plan is executed. It is stored in the "Lot" sub-directory created to store the measured data. The filename given to the Characterization Report is WPROTestPlanRep~<BenchName>.txt.

Where, BenchName is the name of the bench selected for the measurement.

Parametric Test Results Files

Parameteric test results are stored in a .csv for each routine/setup executed in the test plan. They are stored in the "Lot" sub-directory created to store the measured data. This file contains Test Plan Project ID information, the Routine name, the Measurement Condition name and description and finally the Parametric Test Results the measured on each wafer/block/die/temperature/subsite/device selected for characterization as well as the execution status and any warnings or error messages.

The naming convention for these files is:

```
<Bench>~<Mode>~<DeviceType>~<Routine>~<MeasCond>.csv
```

Where,

Bench is the name of the Test Bench selected for the measurement.

Mode is the testplan execution mode

DeviceType is the type of device being tested (MOS, BIP etc).

Routine is the name of the Routine used for the measurement.

MeasCond is the selected Measurement Condition used.

Example

```
OffLine~Simu~MOS~MOS_DC~VDD18.csv
```

Test Plan Time Log File

The Test Plan Time Log File (WProTestPlanTimeLog.txt), is stored in the "Lot" sub-directory created to store the measured data. It contains the time stamps for Test Plan execution along with some project information as well Bench, Run Mode, Wafer, and Temperature information.

Example

```
<Aug 10 Tue 11:59:53 2010> <Project="AgilentExample01"> <Lot="LotSample">
<Bench="OffLine"> <runMode="Simulation"> <TestPlanStart>
<Aug 10 Tue 11:59:53 2010> <SeqID="1"> <Wafer="Wafer_1"> <Temp="25">
<WaferStart>
<Aug 10 Tue 12:03:49 2010> <SeqID="1"> <Wafer="Wafer_1"> <Temp="25"> <WaferEnd>
<Aug 10 Tue 12:03:49 2010> <Project="AgilentExample01"> <Lot="LotSample">
<Bench="OffLine"> <runMode="Simulation"> <TestPlanEnd> <OK>
```

MDM Summary Files

The MDM Summary files, named MdmList_<DeviceType>.csv and stored in the "Lot" sub-directory created to store the measured data contains a spreadsheet table with a summary for .mdm output file created during Test Plan execution.